# Robust Counter-Cascade Defense against Opinion-Conflict Attacks in Online Social Networks

Tianyou Gao
*Kyoto University*
Kyoto, Japan
gao.tianyou.66z@st.kyoto-u.ac.jp

Takayuki Ito
*Kyoto University*
Kyoto, Japan
ito@i.kyoto-u.ac.jp

***Abstract*—Opinion conflict on social networks can be intensified by adversarial information campaigns, while defensive interventions may need to be deployed before the attacker's targets are known. We introduce a Stackelberg counter-cascade model in which a depolarizing defense and a polarizing attack propagate stochastically and subsequently affect users' opinions through Friedkin–Johnsen dynamics. Based on this model, we formulate the Robust Counter-cascade Defense problem, which minimizes the worst-case expected polarization or disagreement against an adaptive attacker. To solve this problem, we develop *Certified Robust Defense (CRD)*, a constraint-generation framework that alternates between certified defense and attack oracles. We derive monotone submodular surrogates for the quadratic opinion-conflict objective and develop efficient attack and defense oracles based on reverse influence sampling. We prove that, with probability at least $1-\delta$, *CRD* certifies an $\varepsilon$-near-optimal defense whenever its certificate gap is at most $\varepsilon$. We further propose *CRD*$^+$, a practical heuristic variant of *CRD* for improved scalability. Experiments on 23 real-world networks demonstrate the effectiveness of both methods, with *CRD*$^+$ scaling to networks with hundreds of millions of edges.**

***Index Terms*—influence maximization, opinion dynamics, robust optimization, counter-cascade defense, polarization**

## I. Introduction

Why can a clarification posted on social media become the starting point of another round of conflict so quickly? Once released, it may be selectively quoted, reframed through a divisive narrative, and recirculated to audiences most likely to react. Prior studies have shown that false information can spread rapidly on social media [1] and that divisive campaigns targeting racial or out-group identities can attract substantial engagement [2], [3]. This phenomenon may reflect coordinated manipulation rather than spontaneous interaction, and such manipulation is far from hypothetical. A report by the U.S. Senate Select Committee on Intelligence documented coordinated social-media operations that exploited racial, ideological, and other societal divisions to stoke anger and push communities further apart [4]. These documented campaigns raise a corresponding defensive question:

> *How should a public agency deploy limited corrective messages when an adversary can adapt its targeting?*

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Consider a defender that can disseminate clarifying or de-escalating messages through only a small number of trusted accounts. The defender must choose where to intervene first, whereas a strategic adversary can observe the deployed intervention and seed divisive content in parts of the network less covered by the defense. Consequently, a defense designed against a fixed attack may become ineffective once the attacker adapts its seed selection.

This strategic setting is closely related to influence maximization (IM) in online social networks (OSNs). Classical IM studies how to select a small set of seed users to maximize the expected spread of a cascade [5]. Competitive IM and misinformation mitigation extend IM to competing cascades, typically selecting seed users to promote a preferred cascade or mitigate the spread and impact of an undesirable one [6]–[9]. However, these formulations generally focus on cascade reach and do not explicitly model how competing exposures reshape users' opinions and affect opinion conflict. More recent cascade-opinion models incorporate post-diffusion opinion dynamics, but typically consider only a single strategically controlled cascade [10], [11].

Against this backdrop, we introduce a Stackelberg counter-cascade model in which defensive and adversarial cascades exert opposing effects on users' opinions. The defender first selects a seed set for a depolarizing counter-cascade, after which the attacker observes the defense and selects a seed set for a polarizing cascade. Cascade exposure first perturbs users' innate opinions, where the attack pushes exposed users away from neutrality and the counter-cascade pulls them toward neutrality while suppressing the adversarial effect. The Friedkin–Johnsen (FJ) model then maps the perturbed innate opinions to equilibrium expressed opinions [12]. We measure the resulting opinion conflict by polarization or disagreement [13]. Figure 1 illustrates the Stackelberg counter-cascade model.

Based on the Stackelberg counter-cascade model, we formulate the *Robust Counter-cascade Defense (RCD)* problem, which seeks a defense minimizing the worst-case expected post-diffusion opinion conflict against an adaptive attacker. To the best of our knowledge, RCD is the first defender-first counter-cascade formulation to explicitly incorporate post-diffusion opinion dynamics. This setting captures interventions in which a platform or public agency may need to commit before the adversary's targets are known.

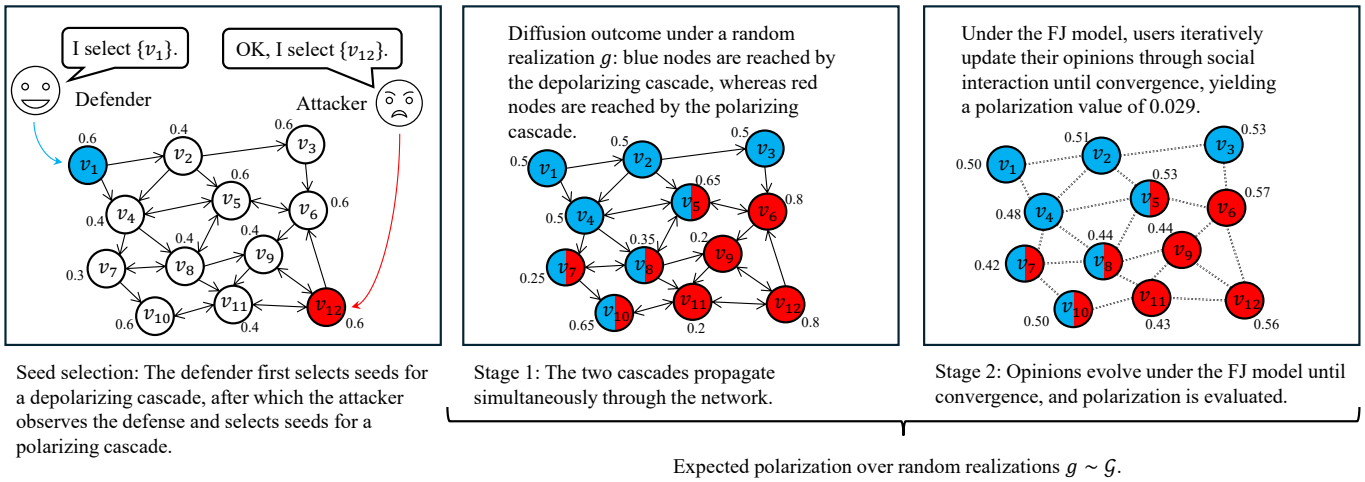



Fig. 1: Illustration of the Stackelberg counter-cascade model. The opinion value of each node is shown beside it, where $0.5$ represents neutrality and values closer to $0$ or $1$ indicate stronger support for opposing viewpoints. The polarizing cascade moves affected opinions away from $0.5$, whereas the depolarizing counter-cascade moves them toward $0.5$.

RCD poses three computational challenges. First, it is a robust min–max optimization problem over two combinatorial seed spaces. Even when one player's seed set is fixed, optimizing the other player's best response remains NP-hard [5]. Second, for any fixed attack–defense pair, exactly evaluating the expected objective is #P-hard in general, as it requires aggregating over exponentially many stochastic propagation realizations and subsumes exact influence-spread computation [14]. Third, both post-diffusion conflict objectives, i.e., polarization and disagreement, are quadratic functions of the equilibrium opinions, coupling local cascade-reachability events through global interactions on the social graph. This coupling prevents the original objective from directly exhibiting the monotone submodular structure exploited by standard IM algorithms [5], [15].

To address these challenges, we develop the *Certified Robust Defense (CRD)* framework. *CRD* iteratively optimizes defenses against a restricted attack pool and invokes an attack oracle to generate candidate attack responses, drawing on cutting-set methods [16], column-and-constraint generation [17], and oracle-based adversarial optimization [18]. Its distinguishing feature is a set of problem-specific probabilistic certificates for RCD. Specifically, *CRD* alternates between a defense oracle and an attack oracle. The defense oracle optimizes against the current pool and returns a candidate defense $S_B$ together with a lower certificate $L$ for the robust optimum. The attack oracle uses an upper surrogate to search the full feasible attack space against $S_B$, returning a candidate attack and an upper certificate $U$ for the worst-case value of $S_B$. Together, $L$ lower-bounds the robust optimum, whereas $U$ upper-bounds the worst-case value of the current defense, which in turn upper-bounds the optimum. Once $U - L \leq \varepsilon$, *CRD* certifies the returned defense as $\varepsilon$-near-optimal with high probability.

The main technical challenge is that the quadratic conflict objective couples individual and pairwise simultaneous-reachability events and does not directly exhibit the monotone submodular structure required by standard IM techniques. Our key step is a realization-wise decomposition of the objective into a constant term, first-order reachability terms, and second-order simultaneous-reachability terms. By bounding these terms in opposite directions, we construct an attacker-side upper surrogate and defender-side lower surrogates. These surrogates are monotone and submodular, enabling weighted reverse influence sampling (RIS) and greedy seed selection with the classical $1 - 1/e$ approximation guarantee [15], [19], which we further use to derive valid upper and lower certificates. Finally, *CRD* accounts for sampling errors and allocates the failure probability across oracle calls, ensuring simultaneous certificate validity with probability at least $1-\delta$. We further propose *CRD*$^+$, a practical variant that uses a linearized objective to improve scalability on large networks.

We conduct extensive experiments on 23 real-world networks, ranging from small benchmark graphs to networks with hundreds of millions of edges. Specifically, our linearized variant *CRD*$^+$ achieves up to $13.7\%$ lower final polarization than the best-performing baseline under the default setting. It also substantially reduces running time relative to *CRD* on the datasets where both methods are evaluated.

In summary, we make the following contributions.

- We introduce a new Stackelberg counter-cascade model that couples opposing stochastic cascades with FJ opinion dynamics, and formulate the RCD problem for minimizing worst-case expected opinion conflict.
- We develop a realization-wise decomposition of the quadratic opinion-conflict objective and derive monotone submodular surrogates that yield valid lower and upper certificates for RCD.
- We propose CRD, a novel defense framework that integrates the proposed certificates with constraint generation

and weighted RIS, and prove that, with probability at least $1-\delta$, it certifies an $\varepsilon$-near-optimal defense whenever its certificate gap is at most $\varepsilon$.
- Experiments on 23 real-world networks demonstrate the defense effectiveness of CRD and its linearized variant *CRD*$^+$, with the latter scaling to networks with hundreds of millions of edges.

A full version of this paper, including additional proofs and experimental results, is available at https://anonymous.4open.science/r/crd/full-version.pdf.

## II. Related Work

### A. Information Diffusion and Influence Maximization

Information diffusion in social networks has been extensively studied in the context of viral marketing and influence propagation [20], [21]. Kempe et al. [5] formulated the classical IM problem under the Independent Cascade (IC) and Linear Threshold (LT) models, and established the monotonicity and submodularity of the expected influence spread, which enables a greedy $1-1/e$ approximation. A large body of subsequent work has improved the efficiency and scalability of IM through lazy evaluation [22], [23], heuristic methods [24], local or path-based approximations [14], [25], [26], and data-driven techniques [27].

A major advance in scalable IM is the RIS technique, introduced by Borgs et al. [28], which estimates influence through randomly sampled reverse reachable sets (RR-sets) and reduces seed selection over the sampled RR-sets to a maximum coverage problem. This technique has led to a series of increasingly efficient algorithms with provable approximation guarantees and improved sampling and time complexity [19], [29]–[32]. The state-of-the-art RIS-based algorithm for classical IM is Hit-and-Stop (*HIST*), proposed by Guo et al. [33], [34], which accelerates RR-set generation through tighter bounds and the hit-and-stop strategy, substantially improving practical efficiency on large-scale networks.

Beyond the classical IM setting, several variants have been studied to address different practical considerations. Robust IM accounts for uncertainty in diffusion parameters or models [35]–[38]. Adaptive IM sequentially selects seeds based on observed diffusion outcomes [39]–[42]. Content-aware IM models the dependence of influence propagation on content characteristics such as topics [43]–[45]. Dynamic IM considers evolving network structures or diffusion processes [37], [46], [47]. Budgeted IM incorporates heterogeneous seed costs or more general resource constraints [42], [48], [49]. These variants address dimensions orthogonal to RCD and are complementary to our formulation.

### B. Adversarial Influence Maximization

Adversarial IM extends classical IM to settings with strategic opponents. Early studies considered competitive IM under competing cascades [6], [50], followed by influence blocking [51], competitive viral marketing [52], more general competitive objectives [53], and richer user behaviors [54]. A particularly relevant branch is misinformation mitigation, where a corrective cascade is deployed to reduce the impact of harmful information [7], [51]. Subsequent studies have incorporated temporal propagation effects [8], fairness constraints [55], and learning-based debunker selection [9].

Beyond competition at the diffusion level, some studies explicitly model sequential decisions between strategic players [6], [50], [56]–[59]. Defender-leading formulations [57], [58] are particularly close to RCD, as the adversary responds to a prior defensive decision. Unlike all the above diffusion-centric formulations, RCD further incorporates post-diffusion opinion dynamics and minimizes the resulting worst-case opinion conflict.

### C. Opinion Dynamics

Opinion dynamics models describe how individuals update their opinions through social interactions. Classical models include the DeGroot model [60] and the FJ model [12], with subsequent studies analyzing opinion formation and disagreement in networked settings [61], [62]. Opinion optimization has also been extensively studied, including opinion maximization [63], [64], controversy reduction [65], and the minimization of polarization and disagreement under FJ dynamics [13], [66].

More recent studies have coupled information diffusion with opinion dynamics [10], [11], [67], [68]. In particular, Tu et al. [10] integrate stochastic information diffusion with FJ opinion dynamics and study seed selection based on the resulting opinion outcomes. Mao et al. [11] further study adversarial perturbation maximization, where diffusion-induced opinion perturbations are optimized to increase polarization or disagreement. However, both studies consider single-cascade or single-sided intervention settings, whereas RCD considers opposing cascades under an adaptive attacker.

## III. Problem Formulation

### A. Preliminaries

*1) Information Propagation Models:* Let $G=(V,E)$ be a directed social graph, where $V$ is the set of users and $E$ is the set of directed social ties. A directed edge $(u,v)\in E$ indicates that user $u$ may activate user $v$ during an information cascade. Let $n=|V|$ and $m=|E|$. We adopt the IC and LT models as standard stochastic information propagation models [5]. Under the IC model, each edge $(u,v)\in E$ is associated with an activation probability $p_{uv}\in[0,1]$, and is independently designated as live with probability $p_{uv}$. Under the LT model, each edge $(u,v)\in E$ is associated with an influence weight $b_{uv}\geq 0$, where $\sum_{u:(u,v)\in E} b_{uv}\leq 1$ for each user $v$. We use its equivalent triggering representation, under which each user $v$ independently selects at most one incoming edge $(u,v)$ as live with probability $b_{uv}$ and selects no incoming edge with the remaining probability. A realization $g$ specifies all live edges under the corresponding model, and the users activated by a seed set are exactly those reachable from the seeds in $g$.

*2) Friedkin–Johnsen Opinion Dynamics:* We use the FJ model as the opinion dynamics model [12]. Following two-stage cascade-opinion models [10], [11], we define the FJ dynamics on the underlying undirected weighted graph of $G$.

For each undirected social tie $\{u, v\}$, let $w_{uv} = w_{vu} \geq 0$ denote the influence weight between users $u$ and $v$. Let $W = (w_{uv})_{u,v \in V}$ be the weighted adjacency matrix of the underlying undirected graph, and let $\Delta = \mathrm{diag}(d_u)_{u \in V}$ with $d_u = \sum_{v \in N(u)} w_{uv}$. The weighted graph Laplacian is $L = \Delta - W$.

Each user $u \in V$ has an innate opinion $x_u \in [0, 1]$, which represents the user's internal belief. Given an innate opinion vector $x \in [0, 1]^n$, the FJ model maps $x$ to the equilibrium expressed opinion vector $z^*(x) = Qx$, where $Q = (I + L)^{-1}$.

*3) Conflict Measures:* We consider two standard conflict measures for the equilibrium expressed opinions, i.e., polarization and disagreement [13]. The polarization measures how far the equilibrium opinions deviate from their mean. Let $\bar{z}(x) = \mathbf{1}^\top z^*(x)/n$. The polarization is $\mathrm{Pol}(x) = \sum_{u \in V}(z^*_u(x) - \bar{z}(x))^2 = x^\top P x$, where $P = Q(I - \mathbf{1}\mathbf{1}^\top/n)Q$. The disagreement measures the weighted opinion difference across social ties. It is $\mathrm{Dis}(x) = \sum_{\{u,v\}} w_{uv}(z^*_u(x) - z^*_v(x))^2 = x^\top D x$, where the summation is over the underlying undirected social ties and $D = QLQ$. We write $C \in \{P, D\}$ when the discussion applies to either conflict measure.

### B. The Stackelberg Counter-Cascade Model

We introduce a Stackelberg counter-cascade model with two strategic cascades that exert opposing effects on users' opinions. The defender first selects a seed set $S_B \subseteq V$ of size $k_B$ for a depolarizing counter-cascade, after which the attacker observes the defense and selects a seed set $S_A \subseteq V$ of size $k_A$ for a polarizing cascade. Both cascades follow a stochastic propagation model, e.g., IC or LT, under the possible-world semantics. Cascade exposure jointly perturbs users' innate opinions, which are then mapped to equilibrium expressed opinions through FJ dynamics. Unlike prior single-cascade [10] or attacker-only models [11], our model explicitly captures strategic interaction between the defense and attack cascades.

Given $g$, the users reached by a cascade from a seed set are exactly those reachable from the seed set through these live edges. The two cascades propagate over the same realization $g$ and do not block each other during diffusion. Thus, a user may be reached by both cascades, and their interaction is modeled only through the subsequent opinion-perturbation rule. For a realization $g$, let $\Gamma(S_A, g)$ and $\Gamma(S_B, g)$ denote the sets of users reached by the attacker and defender cascades, respectively. For each user $u \in V$, let $r^A_u(S_A, g) = \mathbf{1}[u \in \Gamma(S_A, g)]$ and $r^B_u(S_B, g) = \mathbf{1}[u \in \Gamma(S_B, g)]$, which indicate whether user $u$ is exposed to the attacker and defender cascades, respectively, under realization $g$.

Let $s_u \in [0, 1]$ be the original innate opinion of user $u$, and let $s \in [0, 1]^n$ be the original innate opinion vector. Values close to 0 and 1 represent two opposing extremes, while $1/2$ represents a neutral position. Following the two-stage cascade-opinion framework [10], cascade exposure directly perturbs users' innate opinions before the FJ dynamics takes place. We design the perturbation rule according to three principles. First, a polarizing cascade should push a reached user away from the neutral point $1/2$. The signed direction away from neutrality is captured by $s_u - 1/2$, which is negative when $s_u < 1/2$, positive when $s_u > 1/2$, and zero when $s_u = 1/2$. Accordingly, the polarizing cascade does not push all users toward a common target opinion, but instead amplifies their pre-existing deviations from neutrality. In particular, a user with $s_u = 1/2$ is unaffected by the polarizing term. Second, a depolarizing counter-cascade should pull a reached user toward the neutral point, whose signed direction is $1/2 - s_u$. Third, when a user is reached by both cascades, the defensive exposure should not only exert its own depolarizing pull but also suppress the adversarial polarizing effect.

Then, given seed sets $S_A$ and $S_B$ and a realization $g$, we model the joint effect of the two cascades on user $u$ as

$$\hat{s}_u(S_A, S_B, g) = \Pi_{[0,1]}\Bigg[s_u + r^A_u(S_A, g)(1 - \rho_u r^B_u(S_B, g))\alpha_A \cdot \left(s_u - \frac{1}{2}\right) + r^B_u(S_B, g)\alpha_B\left(\frac{1}{2} - s_u\right)\Bigg].$$

Here, $\alpha_A \geq 0$ is the strength of the polarizing cascade, $\alpha_B \geq 0$ is the strength of the depolarizing counter-cascade, $\rho_u \in [0, 1]$ controls how much defensive exposure suppresses the adversarial effect, and $\Pi_{[0,1]}(y) = \min\{1, \max\{0, y\}\}$ clips a value into the interval $[0, 1]$. Let $\hat{s}(S_A, S_B, g) = (\hat{s}_u(S_A, S_B, g))_{u \in V}$ denote the perturbed innate opinion vector under seed sets $S_A, S_B$ and possible world $g$. For notational simplicity, when $S_A$ and $S_B$ are clear from context, we write $\hat{s}(g)$ and $\hat{s}_u(g)$ for $\hat{s}(S_A, S_B, g)$ and $\hat{s}_u(S_A, S_B, g)$, respectively.

### C. Robust Counter-Cascade Defense

We formulate RCD under the Stackelberg counter-cascade model, with both players selecting seeds before stochastic propagation is realized. We fix $|S_B| = k_B$ and $|S_A| = k_A$ to compare defense methods at the same deployment size. The attacker maximizes expected post-propagation conflict, while the defender minimizes its worst-case value. The framework naturally extends to at-most-budget constraints through no-op choices for unused seed slots; these neither activate users nor modify the opinion graph, preserving certificate validity.

For a conflict matrix $C \in \{P, D\}$, define the expected conflict under seed sets $S_A$ and $S_B$ as

$$F_C(S_A, S_B) = \mathbb{E}_{g \sim \mathcal{G}}\left[\hat{s}(S_A, S_B, g)^\top C \hat{s}(S_A, S_B, g)\right], \quad (1)$$

where $\mathcal{G}$ denotes the distribution over realizations induced by the stochastic propagation process. When the conflict matrix is clear from the context, we omit the subscript $C$. Given a defense seed set $S_B$, the attacker selects a best-response seed set $S^*_A$ under budget $k_A$ as

$$S^*_A \in \arg\max_{S_A \subseteq V,\ |S_A| = k_A} F(S_A, S_B).$$

The defender anticipates this best response and selects a seed set $S^*_B$ under budget $k_B$ to minimize the resulting worst-case conflict. Formally, the RCD problem is

$$S^*_B \in \arg\min_{S_B \subseteq V,\ |S_B| = k_B} \max_{S_A \subseteq V,\ |S_A| = k_A} F(S_A, S_B). \quad (2)$$

The inner maximization captures the strongest attack against a fixed defense, and the outer minimization captures the defender's robust choice against this attack.

### D. Computational Challenges

The RCD problem is computationally challenging for three reasons. First, Eq. (2) is a robust min-max optimization problem over two combinatorial seed spaces. Even for a fixed defense $S_B$, the attacker needs to solve a seed-selection problem over all feasible $S_A$, and the defender must further optimize against the resulting worst-case attack. This seed-selection problem subsumes classical IM and is therefore NP-hard [5]. Second, the objective value $F(S_A, S_B)$ is an expectation over stochastic realizations. Thus, exact evaluation requires aggregating the quadratic post-diffusion conflict over exponentially many possible worlds, and inherits the #P-hardness of exact influence computation [14]. Third, the conflict objective is evaluated after the FJ opinion dynamics and takes the quadratic form $\hat{s}(g)^\top C\hat{s}(g)$, which couples the cascade outcome with global opinion interactions on the underlying social graph. These challenges make exact RCD optimization intractable at scale.

## IV. Method

### A. Overview

To address the computational challenges discussed above, *CRD* adopts a constraint-generation strategy. It maintains a finite attack pool $\mathcal{A}$, which contains attack seed sets generated in previous iterations. Given $\mathcal{A}$, the defense oracle computes a candidate defense and a lower certificate for the restricted problem induced by $\mathcal{A}$, while the attack oracle searches over the full feasible attack space for a strong response and computes an upper certificate. The newly generated attack is added to $\mathcal{A}$, and the process is repeated until the certificate gap becomes sufficiently small or the iteration limit is reached.

For a defense seed set $S_B$, define the worst-case attack value (from the defender's perspective, equivalently the attacker's best-response value) as

$$\Phi(S_B) = \max_{S_A \subseteq V,\ |S_A|=k_A} F(S_A, S_B). \tag{3}$$

Then, the optimal value of RCD is $\mathrm{OPT} = \min_{S_B \subseteq V,\ |S_B|=k_B} \Phi(S_B)$. Given an attack pool $\mathcal{A}$, define the restricted attack value as

$$\Phi_{\mathcal{A}}(S_B) = \max_{S_A \in \mathcal{A}} F(S_A, S_B). \tag{4}$$

The corresponding restricted robust defense value is $\mathrm{OPT}_{\mathcal{A}} = \min_{S_B \subseteq V,\ |S_B|=k_B} \Phi_{\mathcal{A}}(S_B)$. Since $\mathcal{A}$ is a subset of the full feasible attack family, we have $\Phi_{\mathcal{A}}(S_B) \le \Phi(S_B)$ for any feasible defense $S_B$, and hence $\mathrm{OPT}_{\mathcal{A}} \le \mathrm{OPT}$.

At iteration $t$, the defense oracle returns a candidate defense $S_B^t$ and a lower certificate $L_t$ with $L_t \le \mathrm{OPT}_{\mathcal{A}} \le \mathrm{OPT}$, while the attack oracle returns an attack seed set $S_A^t$ and an upper certificate $U_t$ with $\Phi(S_B^t) \le U_t$. Since $S_B^t$ is feasible, *CRD* maintains $L_t \le \mathrm{OPT} \le \Phi(S_B^t) \le U_t$. If $U_t - L_t \le \varepsilon$, then $S_B^t$ is certified to be $\varepsilon$-near-optimal. The certificates are computed from weighted RR-set samples, and the sampling errors are incorporated into $L_t$ and $U_t$.

### B. Objective Decomposition

We show that, after fixing one player's seed set, the realized conflict objective can be decomposed into first-order and second-order reachability terms. This decomposition is the common basis for the attacker-side upper certificate and the defender-side lower certificate.

Let $\tau \in \{A, B\}$ denote the player whose seed set is being optimized, and let $\bar{\tau}$ denote the opponent with a fixed seed set. We use $S_\tau$ to denote the optimized seed set and $S_{\bar{\tau}}$ to denote the fixed opponent seed set. When $\tau = A$, we have $S_\tau = S_A$ and $S_{\bar{\tau}} = S_B$. When $\tau = B$, we have $S_\tau = S_B$ and $S_{\bar{\tau}} = S_A$.

Fix an opponent seed set $S_{\bar{\tau}}$ and a realization $g$. The opponent's reachability indicators are fixed under $g$. Thus, the remaining dependence on the optimized seed set $S_\tau$ is through the reachability indicators $r_u^\tau(S_\tau, g)$. For notational simplicity, throughout the rest of this section, we suppress the arguments of $r_u^\tau(S_\tau, g)$, $\eta_u^\tau(g)$, $\xi_u^\tau(g)$, $\ell_u^\tau(g)$, $M_{uv}^\tau(g)$, $\hat{s}_u(S_A, S_B, g)$, and $\hat{s}(S_A, S_B, g)$, whenever they are clear from context.

For each user $u$, we define $\eta_u^\tau$ as the perturbed opinion when user $u$ is not reached by player $\tau$, and $\xi_u^\tau$ as the marginal change when user $u$ is reached by player $\tau$. Since $r_u^\tau \in \{0, 1\}$, the perturbed opinion can be written exactly as $\hat{s}_u = \eta_u^\tau + \xi_u^\tau r_u^\tau$.

When the attacker is optimized against a fixed defense $S_B$, the defender's reachability indicator $r_u^B$ is fixed under $g$. If user $u$ is not reached by the attacker, define

$$\eta_u^A = \Pi_{[0,1]}\left[s_u + r_u^B \alpha_B \left(\frac{1}{2} - s_u\right)\right].$$

The attacker's marginal exposure effect is

$$\xi_u^A = \Pi_{[0,1]}\Big[s_u + (1 - \rho_u r_u^B)\alpha_A \left(s_u - \frac{1}{2}\right) + r_u^B \alpha_B \left(\frac{1}{2} - s_u\right)\Big] - \eta_u^A.$$

Then $\hat{s}_u = \eta_u^A + \xi_u^A r_u^A$.

When the defender is optimized against a fixed attack $S_A$, the attacker's reachability indicator $r_u^A$ is fixed under $g$. If user $u$ is not reached by the defender, define

$$\eta_u^B = \Pi_{[0,1]}\left[s_u + r_u^A \alpha_A \left(s_u - \frac{1}{2}\right)\right].$$

The defender's marginal exposure effect is

$$\xi_u^B = \Pi_{[0,1]}\Big[s_u + r_u^A (1 - \rho_u)\alpha_A \left(s_u - \frac{1}{2}\right) + \alpha_B \left(\frac{1}{2} - s_u\right)\Big] - \eta_u^B.$$

Then $\hat{s}_u = \eta_u^B + \xi_u^B r_u^B$.

For either player $\tau$, let $\eta^\tau = (\eta_u^\tau)_{u \in V}$, $\xi^\tau = (\xi_u^\tau)_{u \in V}$, and $r^\tau = (r_u^\tau)_{u \in V}$. The above definitions imply the vector form $\hat{s} = \eta^\tau + \xi^\tau \odot r^\tau$. Substituting this form into the quadratic conflict objective gives

$$\hat{s}^\top C \hat{s} = (\eta^\tau)^\top C \eta^\tau + \sum_{u \in V} \ell_u^\tau r_u^\tau + \sum_{u,v \in V} M_{uv}^\tau r_u^\tau r_v^\tau. \tag{5}$$

Here, $\ell_u^\tau = 2\xi_u^\tau (C\eta^\tau)_u$ and $M_{uv}^\tau = C_{uv}\xi_u^\tau \xi_v^\tau$.

Eq. (5) decomposes the realized conditional conflict into a constant term, first-order reachability terms, and second-order reachability terms. The constant term does not depend on the optimized seed set $S_\tau$. The first-order terms depend on individual reachability events under player $\tau$'s cascade, while the second-order terms depend on pairwise simultaneous reachability events.

Eq. (5) also gives a direct empirical way to evaluate the original objective $F$ defined in Eq. (1). Inspired by the sampling technique based on pairs of RR-sets for quadratic opinion objectives [10], for fixed seed sets $S_A$ and $S_B$, we may sample a realization $g$, sample an ordered root pair $(u, v)$ uniformly from $V \times V$, and generate the corresponding RR-sets for the two roots under the sampled realization. These RR-sets determine the reachability indicators of $u$ and $v$ under the attacker and defender cascades, and hence determine the perturbed root opinions $\hat{s}_u$ and $\hat{s}_v$. Then $n^2 C_{uv}\hat{s}_u\hat{s}_v$ is an unbiased single-sample estimator of $F(S_A, S_B)$. Averaging independent copies gives an empirical estimator of the original conflict objective for fixed seed sets. However, the corresponding empirical optimization problems do not necessarily preserve monotonicity or submodularity. Therefore, we use this empirical evaluation only as a practical heuristic for generating seed sets. All certificates in our framework are instead derived from the surrogate-based upper and lower bounds introduced below.

*Remark* 1. The quantity $(C\eta^\tau)_u$ need not be computed by explicitly constructing the generally dense conflict matrix $C$. Following the standard treatment of quadratic FJ opinion objectives [10], the product $C\eta^\tau$ can be evaluated through linear-system solves involving $I + L$, exploiting the Laplacian structure of the underlying social graph. Thus, the root-dependent coefficient $\ell_u^\tau = 2\xi_u^\tau (C\eta^\tau)_u$ can be obtained from the corresponding component of this product.

We also use the following matrix bound throughout the subsequent certification analysis.

**Lemma 1.** *For both conflict matrices $C \in \{P, D\}$ considered in this paper, we have $0 \preceq C \preceq I$.*

### C. The Attack Oracle

This subsection describes how *CRD* constructs an upper certificate for the worst-case attack value against a fixed defense. Given a defense seed set $S_B$, the attacker searches over the full feasible attack space. Instead of directly maximizing the quadratic objective $F(S_A, S_B)$, the attack oracle maximizes a monotone submodular upper surrogate derived from Eq. (5).

*1) Fixed-Defense Upper Bound:* Fix a defense seed set $S_B$. For compactness, define $K^A(S_B) = \mathbb{E}_{g\sim\mathcal{G}}\left[(\eta^A)^\top C\eta^A\right]$. Let $[y]_+ = \max\{y, 0\}$. For each realization $g$, define the attacker-side upper node weight as $\psi_u^A = [\ell_u^A + (\xi_u^A)^2]_+$. The corresponding attacker-side upper weighted-influence function is

$$\Psi^A(S_A \mid S_B) = \mathbb{E}_{g\sim\mathcal{G}}\left[\sum_{u\in V} \psi_u^A r_u^A\right]. \tag{6}$$

**Algorithm 1:** Attack Oracle

**Input:** defense seed set $S_B$, attack budget $k_A$, sample size $\theta_A$, sampling error $\varepsilon_A$
**Output:** candidate attack $S_A^\circ$ and upper certificate $U(S_B)$

$\mathcal{R}_A \leftarrow \emptyset$;
**for** $i \leftarrow 1$ **to** $\theta_A$ **do**
  Sample a realization $g_i \sim \mathcal{G}$;
  Sample a root node $u_i$ uniformly from $V$;
  Generate the attacker-side RR-set $R_i^A$ rooted at $u_i$;
  Evaluate the root weight $\psi_{u_i}^A$ under $g_i$ and $S_B$;
  Set $\omega_i^A = n\psi_{u_i}^A$;
  $\mathcal{R}_A \leftarrow \mathcal{R}_A \cup \{(R_i^A, \omega_i^A)\}$;
$S_A^\circ \leftarrow \emptyset$;
**for** $j \leftarrow 1$ **to** $k_A$ **do**
  Select $x^\circ \in \arg\max_{x\in V\setminus S_A^\circ} \widehat{\Psi}^A(S_A^\circ \cup \{x\} \mid S_B)$;
  $S_A^\circ \leftarrow S_A^\circ \cup \{x^\circ\}$;
Compute $U(S_B)$ according to Eq. (10);
**return** $S_A^\circ, U(S_B)$;

All quantities inside the expectation are evaluated under the sampled realization $g$ and the fixed defense $S_B$.

**Lemma 2.** *For any defense seed set $S_B$ and any attack seed set $S_A$, we have*

$$F(S_A, S_B) \le K^A(S_B) + \Psi^A(S_A \mid S_B). \tag{7}$$

**Lemma 3.** *For any fixed defense seed set $S_B$, $\Psi^A(S_A \mid S_B)$ is a monotone submodular set function of $S_A$.*

Define the feasible attack family as $\mathcal{S}_A = \{S_A \subseteq V : |S_A| = k_A\}$. Let $\Psi_{\max}^A(S_B) = \max_{S_A\in\mathcal{S}_A} \Psi^A(S_A \mid S_B)$. By Lemma 2, the worst-case attack value satisfies

$$\Phi(S_B) \le K^A(S_B) + \Psi_{\max}^A(S_B). \tag{8}$$

Therefore, it remains to upper-bound $\Psi_{\max}^A(S_B)$.

*Remark* 2. The term $K^A(S_B)$ need not be evaluated exactly. It can be estimated unbiasedly using the same pair-RIS technique adopted for quadratic opinion objectives [10], together with a one-sided concentration bound to obtain a conservative estimate. Since the derivation follows a standard argument, we omit it for brevity. In a sample-based implementation, the corresponding estimation error and failure probability can be incorporated into the oracle-level budgets $\varepsilon_A$ and $\delta_t^A$, respectively.

*2) Upper Attack Search:* For a realization $g$ and a root node $u$, let $R_u^A(g)$ be the attacker-side RR-set rooted at $u$. A weighted RR-set sample consists of a realization $g_i \sim \mathcal{G}$, a root node $u_i$ sampled uniformly from $V$, the RR-set $R_{u_i}^A(g_i)$, and the sample weight $\omega_i^A = n\psi_{u_i}^A$. Given $\theta_A$ independent weighted RR-set samples, define

$$\widehat{\Psi}^A(S_A \mid S_B) = \frac{1}{\theta_A}\sum_{i=1}^{\theta_A} \omega_i^A \mathbf{1}[R_{u_i}^A(g_i) \cap S_A \neq \emptyset]. \tag{9}$$

**Lemma 4.** *For any fixed defense seed set $S_B$ and any attack seed set $S_A$, the estimator in Eq.* (9) *is unbiased. That is,* $\mathbb{E}[\widehat{\Psi}^A(S_A \mid S_B)] = \Psi^A(S_A \mid S_B)$.

Define $\beta = \frac{1}{2}\max_{u\in V}\sum_{v\in V}|C_{uv}|$. For the candidate attack $S_A^\circ$ returned by Algorithm 1, define the upper certificate

$$U(S_B) = K^A(S_B) + \frac{\widehat{\Psi}^A(S_A^\circ \mid S_B)}{1-\frac{1}{e}} + \varepsilon_A, \tag{10}$$

where $\varepsilon_A > 0$ is the attacker-side sampling error.

*Remark* 3. For both $C = P$ and $C = D$, we have $\beta \le 1$. Indeed, $Q = (I+L)^{-1}$ is nonnegative and satisfies $Q\mathbf{1} = \mathbf{1}$, so every row of $Q$ and $Q^2$ is a probability vector. For polarization, $P = Q^2 - \mathbf{1}\mathbf{1}^\top/n$, while for disagreement, $D = QLQ = Q - Q^2$. Thus, every row of $C$ is the difference of two probability vectors and has $\ell_1$-norm at most 2, yielding $\beta \le 1$.

**Lemma 5.** *Fix a defense seed set $S_B$, and let $\varepsilon_A > 0$. If*

$$\theta_A \ge \frac{n^2(2\beta+1)^2}{2\varepsilon_A^2}\log\frac{1}{\delta_1}, \tag{11}$$

*then, with probability at least $1-\delta_1$, the upper certificate $U(S_B)$ returned by Algorithm 1 satisfies $\Phi(S_B) \le U(S_B)$.*

### D. The Defense Oracle

This subsection describes how *CRD* constructs a lower certificate for the robust optimum using the current attack pool. The key idea is to aggregate fixed-attack lower functions over the attack pool. We use the uniform distribution over the attack pool as the default mixed attack distribution. This gives a pooled lower certificate in which a single defense must simultaneously reduce the aggregate lower surrogate induced by all attacks in the pool.

*1) Fixed-Attack Lower Functions:* Fix an attack seed set $S_A$. For compactness, define $K^B(S_A) = \mathbb{E}_{g\sim\mathcal{G}}\left[(\eta^B)^\top C\eta^B\right]$. For each realization $g$, define the defender-side lower node weight as $\psi_u^B = \left[-\ell_u^B\right]_+$. The corresponding defender-side lower weighted-influence function is

$$\Psi^B(S_B \mid S_A) = \mathbb{E}_{g\sim\mathcal{G}}\left[\sum_{u\in V}\psi_u^B r_u^B\right]. \tag{12}$$

Define the fixed-attack lower function as

$$h(S_B \mid S_A) = K^B(S_A) - \Psi^B(S_B \mid S_A). \tag{13}$$

**Lemma 6.** *For any attack seed set $S_A$ and any defense seed set $S_B$, we have*

$$F(S_A, S_B) \ge h(S_B \mid S_A). \tag{14}$$

**Lemma 7.** *For any fixed attack seed set $S_A$, $\Psi^B(S_B \mid S_A)$ is a monotone submodular set function of $S_B$.*

**Algorithm 2:** Lower Defense Search

**Input:** attack pool $\mathcal{A}$, attack distribution $\pi$, defense budget $k_B$, sample size $\theta_B$, sampling error $\varepsilon_B$
**Output:** mixed surrogate defense set $S_B^\circ(\pi)$ and lower certificate $L_{\mathrm{mix}}(\pi)$

$\mathcal{R}_B \leftarrow \emptyset$;
**for** $i \leftarrow 1$ **to** $\theta_B$ **do**
  Sample an attack seed set $T_i$ from $\mathcal{A}$ according to $\pi$;
  Sample a realization $g_i \sim \mathcal{G}$;
  Sample a root node $u_i$ uniformly from $V$;
  Generate the defender-side RR-set $R_i^B$ rooted at $u_i$;
  Evaluate the root weight $\psi_{u_i}^B$ under $g_i$ and $T_i$;
  Set $\omega_i^B = n\psi_{u_i}^B$;
  $\mathcal{R}_B \leftarrow \mathcal{R}_B \cup \{(R_i^B, \omega_i^B)\}$;
$S_B^\circ(\pi) \leftarrow \emptyset$;
**for** $j \leftarrow 1$ **to** $k_B$ **do**
  Select $x^\circ \in \arg\max_{x\in V\setminus S_B^\circ(\pi)} \widehat{\Psi}_\pi^B(S_B^\circ(\pi)\cup\{x\})$;
  $S_B^\circ(\pi) \leftarrow S_B^\circ(\pi)\cup\{x^\circ\}$;
Compute $L_{\mathrm{mix}}(\pi)$ according to Eq. (16);
**return** $S_B^\circ(\pi), L_{\mathrm{mix}}(\pi)$;

*2) Mixed-Attack Lower Bound:* Let $\mathcal{A} = \{S_A^1, \dots, S_A^p\}$ be a nonempty attack pool. Let $\pi$ be a distribution over $\mathcal{A}$, where $\pi_i \ge 0$ and $\sum_{i=1}^p \pi_i = 1$. For each attack $S_A^i$, write $h_i(S_B) = h(S_B \mid S_A^i)$. The mixed lower function induced by $\pi$ is $h_\pi(S_B) = \sum_{i=1}^p \pi_i h_i(S_B)$. Equivalently, we can write $h_\pi(S_B) = K_\pi^B - \Psi_\pi^B(S_B)$, where $K_\pi^B = \sum_{i=1}^p \pi_i K^B(S_A^i)$ and $\Psi_\pi^B(S_B) = \sum_{i=1}^p \pi_i \Psi^B(S_B \mid S_A^i)$. Since each $\Psi^B(\cdot \mid S_A^i)$ is monotone submodular by Lemma 7, the mixed term $\Psi_\pi^B(\cdot)$ is also monotone submodular.

For a realization $g$ and a root node $u$, let $R_u^B(g)$ be the defender-side RR-set rooted at $u$. A mixed weighted RR-set sample consists of an attack $T_q \sim \pi$, a realization $g_q \sim \mathcal{G}$, a root node $u_q$ sampled uniformly from $V$, the defender-side RR-set $R_{u_q}^B(g_q)$, and the sample weight $\omega_q^B = n\psi_{u_q}^B$, where $\psi_{u_q}^B$ is evaluated under the sampled attack $T_q$ and realization $g_q$. Given $\theta_B$ independent mixed weighted RR-set samples, define

$$\widehat{\Psi}_\pi^B(S_B) = \frac{1}{\theta_B}\sum_{q=1}^{\theta_B}\omega_q^B\mathbf{1}[R_{u_q}^B(g_q)\cap S_B \ne \emptyset]. \tag{15}$$

**Lemma 8.** *For any attack distribution $\pi$ and any defense seed set $S_B$, the estimator in Eq.* (15) *is unbiased. That is,* $\mathbb{E}[\widehat{\Psi}_\pi^B(S_B)] = \Psi_\pi^B(S_B)$.

Let $\mathcal{S}_B = \{S_B \subseteq V : |S_B| = k_B\}$, $\Psi_{\max,\pi}^B = \max_{S_B\in\mathcal{S}_B}\Psi_\pi^B(S_B)$. For the set $S_B^\circ(\pi)$ returned by Algorithm 2, define the mixed-attack lower bound as

$$L_{\mathrm{mix}}(\pi) = K_\pi^B - \frac{\widehat{\Psi}_\pi^B(S_B^\circ(\pi))}{1-\frac{1}{e}} - \varepsilon_B, \tag{16}$$

where $\varepsilon_B > 0$ is the defender-side sampling error.

**Algorithm 3:** Empirical Defense Search

**Input:** attack pool $\mathcal{A}$, defense budget $k_B$, empirical sample size $\theta_{\mathrm{emp}}$

**Output:** empirical defense candidate $S'_B$ and empirical pool evaluator $\widehat{\Phi}_{\mathcal{A}}$

Construct an empirical pool evaluator $\widehat{\Phi}_{\mathcal{A}}$ using $\theta_{\mathrm{emp}}$ samples of pairs of RR-sets;
$S'_B \leftarrow \emptyset$;
**for** $j \leftarrow 1$ **to** $k_B$ **do**
  Select $x' \in \arg\min_{x \in V \setminus S'_B} \widehat{\Phi}_{\mathcal{A}}(S'_B \cup \{x\})$;
  $S'_B \leftarrow S'_B \cup \{x'\}$;
**return** $S'_B, \widehat{\Phi}_{\mathcal{A}}$;

*Remark* 4. Similar to Remark 2, $K_\pi^B$ can be estimated unbiasedly using pair-RIS and equipped with a one-sided concentration bound. We omit the routine derivation for brevity.

*3) Lower Defense Search:* We establish the sampling guarantee for the lower defense search. The following lemma gives a sufficient sample size under which $L_{\mathrm{mix}}(\pi)$ is a valid lower certificate with high probability.

**Lemma 9.** *Let $\varepsilon_B > 0$. If*

$$\theta_B \geq \frac{2n^2\beta^2}{\varepsilon_B^2} \log \frac{1}{\delta_2}, \tag{17}$$

*then, with probability at least $1-\delta_2$, the lower bound $L_{\mathrm{mix}}(\pi)$ returned by Algorithm 2 satisfies $L_{\mathrm{mix}}(\pi) \leq \mathrm{OPT}_{\mathcal{A}} \leq \mathrm{OPT}$.*

In the main algorithm, we use the uniform distribution over the attack pool. That is, for $\mathcal{A} = \{S_A^1, \ldots, S_A^p\}$, we set $\pi_i = 1/p$ for every $i \in [p]$. More adaptive choices of $\pi$ can be generated by increasing the weights of attacks whose lower payoffs remain high under the current mixed defense witnesses. We describe this adaptive extension in the full version.

*4) Empirical Defense Search:* The defense oracle works with a finite attack pool $\mathcal{A}$. It returns two objects with different roles. The candidate defense $S_B$ is used by the outer *CRD* algorithm to query the attack oracle. The lower certificate $L$ is used only to certify progress and does not rely on the optimality of the returned candidate defense.

Algorithm 3 constructs an empirical evaluator $\widehat{\Phi}_{\mathcal{A}}$ for ranking defense candidates against the current attack pool. A natural implementation is $\widehat{\Phi}_{\mathcal{A}}(S_B) = \max_{S_A \in \mathcal{A}} \widehat{F}(S_A, S_B)$, where $\widehat{F}$ is obtained from the empirical estimator discussed after Eq. (5). This empirical evaluator is used only as a practical heuristic. It does not necessarily preserve monotonicity or submodularity, and it is not used as a certificate.

*5) Assembling the Defense Oracle:* Algorithm 4 first computes the empirical defense candidate $S'_B$. It then constructs the uniform mixed-attack distribution over $\mathcal{A}$ and calls Algorithm 2 to obtain the surrogate defense set $S_B^\circ(\pi)$ and the lower certificate $L$. The final candidate defense is selected between $S'_B$ and $S_B^\circ(\pi)$ using the empirical evaluator $\widehat{\Phi}_{\mathcal{A}}$.

**Algorithm 4:** Defense Oracle

**Input:** attack pool $\mathcal{A}$, defense budget $k_B$, empirical sample size $\theta_{\mathrm{emp}}$, sample size $\theta_B$, sampling error $\varepsilon_B$

**Output:** candidate defense $S_B$ and lower certificate $L$

$(S'_B, \widehat{\Phi}_{\mathcal{A}}) \leftarrow$ Empirical Defense Search$(\mathcal{A}, k_B, \theta_{\mathrm{emp}})$;
Set $\pi_i = 1/|\mathcal{A}|$ for every $S_A^i \in \mathcal{A}$;
$(S_B^\circ(\pi), L) \leftarrow$ Lower Defense Search$(\mathcal{A}, \pi, k_B, \theta_B, \varepsilon_B)$;
$S_B \leftarrow \arg\min_{S \in \{S'_B, S_B^\circ(\pi)\}} \widehat{\Phi}_{\mathcal{A}}(S)$;
**return** $S_B, L$;

**Algorithm 5:** The Certified Robust Defense Algorithm

**Input:** graph $G$, opinion vector $s$, budgets $k_A, k_B$, conflict matrix $C$, initial attack pool $\mathcal{A}_0$, tolerance $\varepsilon$, failure probability $\delta$, iteration limit $T$

**Output:** defense seed set $S_B$ and certificate interval $[L, U]$

$\mathcal{A} \leftarrow \mathcal{A}_0$;
Set $\varepsilon_A = \varepsilon_B = \varepsilon/4$;
$(S_B^{\mathrm{best}}, L^{\mathrm{best}}, U^{\mathrm{best}}) \leftarrow (\emptyset, -\infty, +\infty)$;
**for** $t \leftarrow 1$ **to** $T$ **do**
  Set $\delta_t^A = \delta_t^B = \delta/[2t(t+1)]$;
  Set $\theta_A^t$ by Eq. (11) with $\varepsilon_A$ and $\delta_1 = \delta_t^A$;
  Set $\theta_B^t$ by Eq. (17) with $\varepsilon_B$ and $\delta_2 = \delta_t^B$;
  $(S_B^t, L_t) \leftarrow$ Defense Oracle$(\mathcal{A}, k_B, \theta_B^t, \theta_B^t, \varepsilon_B)$;
  $(S_A^t, U_t) \leftarrow$ Attack Oracle$(S_B^t, k_A, \theta_A^t, \varepsilon_A)$;
  **if** $U_t - L_t < U^{\mathrm{best}} - L^{\mathrm{best}}$ **then**
    $(S_B^{\mathrm{best}}, L^{\mathrm{best}}, U^{\mathrm{best}}) \leftarrow (S_B^t, L_t, U_t)$;
  **if** $U_t - L_t \leq \varepsilon$ **then**
    **return** $S_B^t, [L_t, U_t]$;
  $\mathcal{A} \leftarrow \mathcal{A} \cup \{S_A^t\}$;
**return** $S_B^{\mathrm{best}}, [L^{\mathrm{best}}, U^{\mathrm{best}}]$;

**Lemma 10.** *Assume that Algorithm 2 returns a valid mixed lower certificate $L_{\mathrm{mix}}(\pi)$. Then the lower certificate $L$ returned by Algorithm 4 satisfies $L \leq \mathrm{OPT}$.*

The validity of $L$ does not depend on the empirical evaluator or on which candidate defense is selected. In sample-based implementations, $K_\pi^B$ is replaced by a lower confidence bound, while the sampling error of $\widehat{\Psi}_\pi^B(S_B^\circ(\pi))$ is controlled by $\varepsilon_B$ as above. The corresponding failure probabilities are allocated so that Lemma 9 holds with the prescribed probability.

### *E. The CRD Algorithm*

Algorithm 5 gives the complete *CRD* framework. *CRD* maintains a finite attack pool and iteratively refines it with newly generated attacks. At iteration $t$, the defense oracle returns a candidate defense $S_B^t$ and a lower certificate $L_t$, while the attack oracle returns an attack seed set $S_A^t$ and an upper certificate $U_t$. If $U_t - L_t \leq \varepsilon$, *CRD* terminates and

returns a certified $\varepsilon$-near-optimal defense. Otherwise, $S_A^t$ is added to the attack pool.

The initial attack pool $\mathcal{A}_0$ is assumed to be nonempty and is initialized by default with the $k_A$ nodes of largest out-degree. The empirical sample size for ranking defense candidates is set to $\theta_B^t$, while the certified sample sizes $\theta_A^t$ and $\theta_B^t$ are determined by Eqs. (11) and (17), respectively. At iteration $t$, we allocate $\delta_t^A = \delta_t^B = \delta/[2t(t+1)]$. Since $\sum_{t=1}^{\infty} 1/[t(t+1)] = 1$, the total failure probability over all oracle calls is at most $\delta$, independently of the iteration limit $T$. Thus, $T$ serves only as a computational safeguard.

The following theorem states the certificate guarantee maintained by *CRD*.

**Theorem 11.** *With probability at least $1-\delta$, for every executed iteration $t$, Algorithm 5 satisfies $L_t \le \mathrm{OPT} \le \Phi(S_B^t) \le U_t$. Consequently, if $U_t - L_t \le \varepsilon$, then the returned defense $S_B^t$ satisfies*

$$\Phi(S_B^t) - \mathrm{OPT} \le \varepsilon. \tag{18}$$

If the iteration limit is reached before the stopping condition is satisfied, Algorithm 5 returns the defense with the smallest observed certificate gap.

*Remark* 5. To improve scalability, we introduce *CRD*$^+$, a natural and practical variant of *CRD*. *CRD*$^+$ differs from *CRD* only in that it omits the second-order term $\sum_{u,v\in V} M_{uv}^{\tau} r_u^{\tau} r_v^{\tau}$ and retains the first-order term $\sum_{u\in V} \ell_u^{\tau} r_u^{\tau}$ in Eq. (5). The resulting linearized objective is optimized using the same weighted-RIS machinery as *CRD*. *CRD*$^+$ does not inherit the certificate guarantee of the full *CRD* framework. Its practical efficiency and solution quality are evaluated in Section V.

### F. Complexity Analysis

By Remark 3, $\beta \le 1$. Under the parameter settings of Algorithm 5, the sample-size bounds in Eqs. (11) and (17) give $\theta_A^t, \theta_B^t = O(n^2\varepsilon^{-2}\log(2t(t+1)/\delta))$. Therefore, over at most $T$ iterations, *CRD* generates $O(Tn^2\varepsilon^{-2}\log(2T(T+1)/\delta))$ weighted RR-set samples.

Let $c_{\mathrm{RR}}$ denote the expected cost of generating and processing one weighted RR-set sample, including its root weight. Since the standard RIS greedy procedure can be implemented in time linear in the total RR-set incidence size [32], [34], the expected running time of *CRD* is $O(Tn^2 c_{\mathrm{RR}}\varepsilon^{-2}\log(2T(T+1)/\delta))$. For constant $T$ and the conservative bound $c_{\mathrm{RR}} = O(n)$, the expected running time becomes $O(n^3\varepsilon^{-2}\log(1/\delta))$. Under the standard RIS implementation, the space complexity has the same asymptotic order as the running-time bound, up to the $O(m)$ space required to store the graph.

## V. Experiments

We conduct experiments to answer the following questions:

- How do the proposed methods compare with representative baselines?
- How consistent is the performance of the proposed methods across different parameter settings and experimental settings?
- How important are the major components of *CRD*?
- How do FJ computation and linearization affect solution quality and efficiency?

TABLE I: Dataset Statistics

| Dataset | $n$ | $m$ | Dataset | $n$ | $m$ |
|---|---|---|---|---|---|
| *Krackhardt* | 21 | 190 | *HepPh* | 11.2K | 235K |
| *Karate* | 34 | 156 | *AnyBeat* | 12.6K | 98.3K |
| *Convote* | 219 | 1.04K | *CondMat* | 21.4K | 183K |
| *NetScience* | 379 | 1.83K | *GPlus* | 23.6K | 78.4K |
| *WikiTalkHT* | 404 | 1.47K | *Brightkite* | 56.7K | 426K |
| *WikiVote* | 889 | 5.83K | *WikiTalk* | 92.1K | 722K |
| *Reed98* | 962 | 37.6K | *Epinions* | 132K | 841K |
| *EmailUniv* | 1.13K | 10.9K | *Gowalla* | 197K | 1.90M |
| *Hamster* | 2.00K | 32.2K | *YouTube* | 1.13M | 5.98M |
| *USFCA72* | 2.67K | 130K | *Pokec* | 1.63M | 30.6M |
| *NIPS-Ego* | 2.89K | 5.96K | *Orkut* | 3.07M | 234M |
| *PagesGov* | 7.06K | 179K | | | |

K and M denote $10^3$ and $10^6$, respectively.

### A. Experimental Setup

All experiments are conducted on a Linux server with 251 GB of memory. We implement all algorithms in C++ and compile them with the `-O3` optimization flag. Our source code is publicly available at https://anonymous.4open.science/r/crd.

*1) Datasets:* We use 23 publicly available real-world networks spanning several orders of magnitude in size. Most datasets are obtained from SNAP [69], Network Repository [70], and KONECT [71]. In addition, *Krackhardt* is obtained from [72], while *WikiTalkHT* is obtained from [73]. Table I summarizes their statistics. We use the two smallest networks, *Krackhardt* and *Karate*, for validation with exhaustive search.

*2) Preprocessing:* We preprocess each dataset to construct the directed social graph $G$ and the corresponding undirected weighted graph for FJ dynamics. For each originally undirected network, every social tie is represented by two oppositely directed edges in $G$, while originally directed networks retain their prescribed edge directions. The underlying undirected weighted graph used for FJ dynamics is constructed by deduplicating unordered endpoint pairs, such that reciprocal directed edges in $G$ correspond to a single undirected social tie. By default, we adopt the Weighted Cascade (WC) setting $p_{uv} = 1/d_v^{\mathrm{in}}$ under IC, where $d_v^{\mathrm{in}}$ denotes the in-degree of node $v$, and assign unit weights to all undirected social ties for FJ dynamics.

*3) Compared Methods:* To the best of our knowledge, no existing method directly addresses the RCD problem. We therefore compare *CRD* and its scalable variant *CRD*$^+$ with representative methods originally designed for related problem settings, which we adapt to the RCD setting to capture complementary aspects of the problem.

- *FJ-Iter* [10]: An opinion-aware iterative baseline that alternates between defense and attack seed selection over multiple rounds using pair-RIS estimates [10] of the FJ conflict objective. Each newly generated attack is used to update the defense in the subsequent round.
- *HIST-Iter* [57]: An iterative extension of *HIST* [34] that alternates between defense selection and attack gener-

ation in a defender-leading manner. It maintains a cumulative attack pool and selects defense seeds based on residual cascade reachability against the accumulated attacks.

- *HIST* [33], [34]: A state-of-the-art RIS-based algorithm for classical IM [5].
- *Degree*: A baseline that selects the nodes with the largest out-degrees as defense seeds.
- *Random*: A baseline that selects defense seeds uniformly without replacement.
- *Exact*: A small-network reference that exhaustively enumerates fixed-cardinality attack and defense seed sets, with each pair evaluated using $10^4$ MC simulations. Notably, the enumeration is exact, whereas the stochastic objective is only estimated. We use *Exact* only for small-scale validation because exhaustive seed set enumeration is infeasible on larger networks.

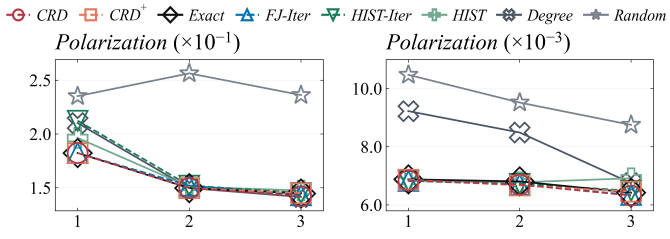
The full *CRD* and *FJ-Iter* are evaluated on the nine datasets in Table I, from *Convote* to *NIPS-Ego*. For the remaining larger datasets, from *PagesGov* onward, we evaluate *CRD*$^+$ together with the scalable baselines.

*4) Default Parameter Settings:* We use polarization as the conflict objective, the IC model for propagation, and fixed innate-opinion vectors with entries sampled uniformly from $[0, 1]$. We set $k_A = k_B = \lceil 0.5\% \cdot n \rceil$, $\alpha_A = \alpha_B = 0.5$, and $\rho_u = 0.5$ for all $u \in V$. For *CRD*, we set $T = 10$, $\varepsilon = 0.01n$, and $\delta = 0.01$. The iterative baselines also use at most 10 outer iterations. The theoretical sample-size bounds provide conservative sufficient conditions for certification and can be costly to satisfy in practice. Therefore, we use a fixed empirical sampling budget of $10^5$ for each of the attack, defense, and empirical-selection components of both *CRD* and *CRD*$^+$, and $10^5$ pair-RR samples for *FJ-Iter*. This trade-off between empirical solution quality and computational cost is examined in Section V-D4. For *HIST* and *HIST-Iter*, the accuracy parameter and failure probability are set to 0.1 and 0.01, respectively.

*5) Evaluation Protocol:* We evaluate each defense against a generated attack response using the full post-cascade FJ conflict objective, i.e., polarization under the default setting. To enable evaluation at scale, we generate the final attack against each defense using Algorithm 1. Applying the same attack-generation procedure across methods provides a consistent basis for comparison while allowing the attacker to adapt to each defense. The resulting defense–attack pair is then independently evaluated using $10^4$ MC simulations of the complete propagation process [29], [32] and subsequent FJ dynamics. We repeat each configuration ten times and report the average results.

### B. Defense Effectiveness

Table II reports the final polarization achieved by the compared methods under the default setting. Across the nine datasets where *CRD* is evaluated, it achieves the lowest polarization on six datasets and remains close to the best baseline on the remaining three. In particular, *CRD* consistently outperforms *FJ-Iter* and remains competitive with *HIST-Iter*, *HIST*, and *Degree*. Meanwhile, *CRD*$^+$ achieves the lowest polarization on all datasets where it is evaluated. Specifically, on *Orkut*, our largest network, *CRD*$^+$ achieves 13.7% lower final polarization than the best-performing baseline, *HIST-Iter*.




(a) *Karate* (b) *Krackhardt*

Fig. 2: Final polarization vs. seed budget on tiny networks.

As expected, *CRD* requires more running time than *FJ-Iter* and *HIST-Iter*. For larger networks, we use the linearized variant *CRD*$^+$, which remains computationally feasible even on our largest network, *Orkut*. Detailed running-time results are provided in the full version.

### C. Validation against Exhaustive Search

We validate the proposed methods on the two small networks, *Karate* and *Krackhardt*, where exhaustive seed set search is feasible. We set $k_A = k_B = k$ and vary $k \in \{1, 2, 3\}$. As shown in Fig. 2, both *CRD* and *CRD*$^+$ consistently match *Exact* and achieve the lowest final polarization across all tested seed budgets. These results provide additional empirical evidence for the solution quality of the proposed methods.

### D. Sensitivity Analysis

We examine sensitivity to the main parameters. Some detailed results are provided in the full version.

*1) Seed Budgets:* We study the sensitivity to the seed budgets by first setting $k_A = k_B = \lceil rn \rceil$ and varying $r$ from 0.05% to 4%. As shown in Fig. 3, the final polarization generally decreases as the symmetric seed budget increases. Across different budget settings, *CRD* and *CRD*$^+$ consistently achieve the lowest or near-lowest polarization, with *CRD*$^+$ outperforming *CRD* on some datasets. We further vary $k_A$ and $k_B$ separately and observe that increasing the attack budget generally increases polarization, whereas increasing the defense budget reduces it. These results are consistent with the opposing roles of the attack and defense cascades.

*2) Cascade Parameters:* We examine the sensitivity to $\rho_u$, $\alpha_A$, and $\alpha_B$, setting $\rho_u = \rho$ for all $u \in V$ when varying $\rho$. The relative performance of the compared methods remains consistent across different settings. Increasing $\rho$ or $\alpha_B$ generally reduces polarization, whereas increasing $\alpha_A$ increases it.

*3) Iteration Limit:* We vary the maximum number of outer iterations $T$. On most datasets, varying $T$ has little effect on the final polarization. This result supports $T = 10$ as a reasonable default setting in our experiments.

TABLE II: Final Polarization under the Default Setting

| Dataset | *CRD*/*CRD*$^+$ | *FJ-Iter* | *HIST-Iter* | *HIST* | *Degree* | *Random* |
|---|---|---|---|---|---|---|
| *Convote* | $\mathbf{2.26 \pm 0.03}$ | $2.33 \pm 0.10$ | $2.29 \pm 0.03$ | $2.28 \pm 0.03$ | $2.27 \pm 0.01$ | $2.74 \pm 0.04$ |
| *NetScience* | $\mathbf{4.90 \pm 0.07}$ | $5.23 \pm 0.16$ | $5.23 \pm 0.05$ | $5.20 \pm 0.02$ | $5.39 \pm 0.01$ | $5.42 \pm 0.08$ |
| *WikiTalkHT* | $\mathbf{3.71 \pm 0.04}$ | $3.77 \pm 0.07$ | $3.78 \pm 0.08$ | $3.78 \pm 0.06$ | $3.77 \pm 0.00$ | $7.08 \pm 0.04$ |
| *WikiVote* | $\mathbf{7.00 \pm 0.24}$ | $8.13 \pm 0.32$ | $7.22 \pm 0.15$ | $7.15 \pm 0.15$ | $7.27 \pm 0.01$ | $8.49 \pm 0.07$ |
| *Reed98* | $1.29 \pm 0.10$ | $1.49 \pm 0.03$ | $1.27 \pm 0.01$ | $1.27 \pm 0.00$ | $\mathbf{1.27 \pm 0.00}$ | $1.49 \pm 0.02$ |
| *EmailUniv* | $\mathbf{5.84 \pm 0.21}$ | $6.27 \pm 0.09$ | $5.87 \pm 0.05$ | $5.86 \pm 0.03$ | $5.92 \pm 0.02$ | $6.41 \pm 0.05$ |
| *Hamster* | $7.43 \pm 0.42$ | $8.21 \pm 0.20$ | $7.42 \pm 0.05$ | $7.42 \pm 0.07$ | $\mathbf{7.41 \pm 0.01}$ | $8.41 \pm 0.06$ |
| *USFCA72* | $3.23 \pm 0.21$ | $3.53 \pm 0.12$ | $3.21 \pm 0.04$ | $3.23 \pm 0.03$ | $\mathbf{3.18 \pm 0.01}$ | $3.62 \pm 0.05$ |
| *NIPS-Ego* | $\mathbf{33.26 \pm 0.00}$ | $65.32 \pm 18.56$ | $33.28 \pm 0.00$ | $36.43 \pm 1.38$ | $33.28 \pm 0.00$ | $97.47 \pm 0.09$ |
| *PagesGov* | $\mathbf{18.32 \pm 0.05}$ | — | $20.59 \pm 0.24$ | $20.60 \pm 0.12$ | $21.66 \pm 0.02$ | $23.61 \pm 0.08$ |
| *HepPh* | $\mathbf{70.99 \pm 0.16}$ | — | $74.75 \pm 0.33$ | $74.42 \pm 0.42$ | $76.92 \pm 0.10$ | $80.40 \pm 0.36$ |
| *AnyBeat* | $\mathbf{115.34 \pm 0.21}$ | — | $121.52 \pm 0.97$ | $129.77 \pm 1.45$ | $122.07 \pm 0.08$ | $246.39 \pm 1.29$ |
| *CondMat* | $\mathbf{134.79 \pm 0.09}$ | — | $143.62 \pm 0.46$ | $142.07 \pm 0.53$ | $144.19 \pm 0.14$ | $153.36 \pm 0.30$ |
| *GPlus* | $\mathbf{225.16 \pm 0.04}$ | — | $260.17 \pm 34.06$ | $285.74 \pm 17.66$ | $226.11 \pm 0.01$ | $636.20 \pm 4.06$ |
| *Brightkite* | $\mathbf{623.32 \pm 0.52}$ | — | $662.32 \pm 4.67$ | $659.06 \pm 4.12$ | $653.24 \pm 0.52$ | $814.12 \pm 3.85$ |
| *WikiTalk* | $\mathbf{1029.21 \pm 1.04}$ | — | $1178.19 \pm 21.62$ | $1192.31 \pm 18.23$ | $1089.17 \pm 0.95$ | $1794.25 \pm 9.39$ |
| *Epinions* | $\mathbf{2021.71 \pm 1.40}$ | — | $2162.91 \pm 3.24$ | $2124.31 \pm 4.94$ | $2100.69 \pm 1.22$ | $2440.50 \pm 5.19$ |
| *Gowalla* | $\mathbf{1740.48 \pm 0.94}$ | — | $1869.24 \pm 3.69$ | $1841.92 \pm 3.94$ | $1788.09 \pm 1.43$ | $2332.67 \pm 3.89$ |
| *YouTube* | $\mathbf{14590.33 \pm 4.70}$ | — | $17561.49 \pm 38.26$ | $17012.68 \pm 47.46$ | $15226.96 \pm 8.98$ | $23110.04 \pm 25.32$ |
| *Pokec* | $\mathbf{5576.08 \pm 4.98}$ | — | $6518.53 \pm 7.77$ | $6332.32 \pm 6.95$ | $5839.92 \pm 4.57$ | $6699.69 \pm 4.75$ |
| *Orkut* | $\mathbf{2245.12 \pm 2.33}$ | — | $2601.23 \pm 2.87$ | $2603.90 \pm 5.26$ | $2642.07 \pm 2.56$ | $2929.93 \pm 3.44$ |

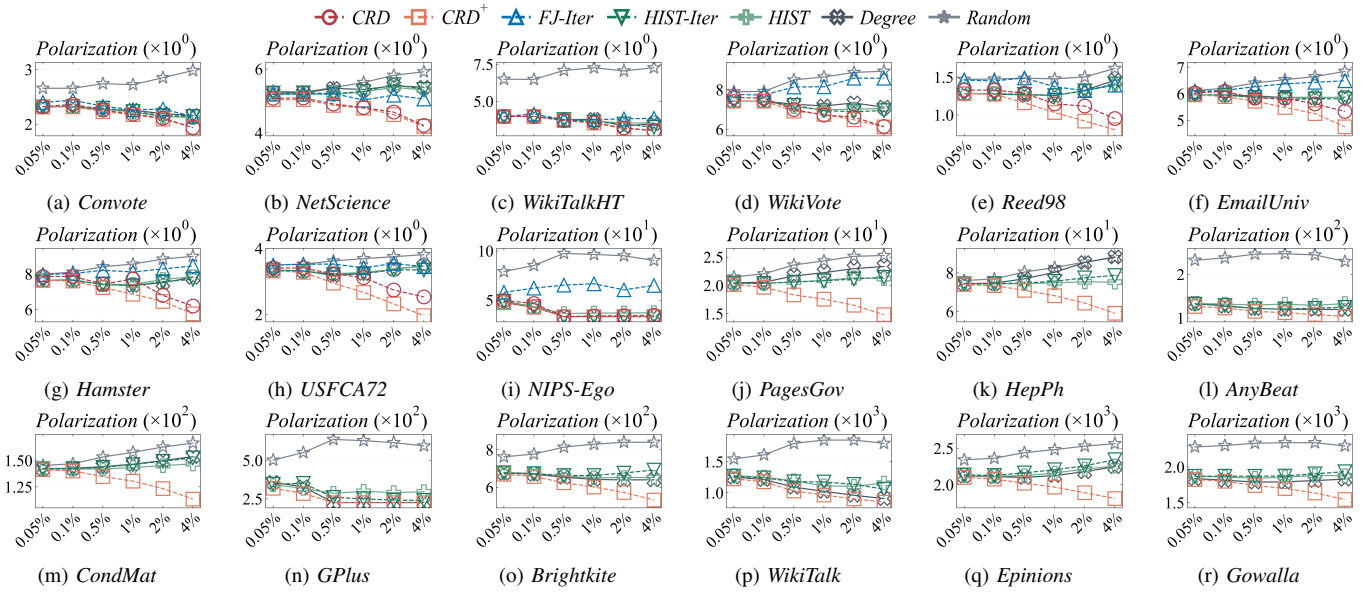



Fig. 3: Final polarization vs. seed budget for different methods.

*4) Sampling Budget:* We vary the sampling budget from $10^3$ to $10^6$. Increasing the sampling budget generally improves or stabilizes the final polarization, but the gains become limited beyond $10^5$ on most datasets, while the running time continues to increase substantially. These results support $10^5$ as a practical default sampling budget that balances solution quality and computational cost.

*5) Certificate Tolerance:* We vary the certificate tolerance $\varepsilon$. Relaxing the tolerance has only a limited effect on the final polarization, but substantially reduces the running time. These results show that the certificate tolerance mainly controls the computational cost, while the resulting solution quality remains relatively stable.

## *E. Generality*

We evaluate the generality of the proposed methods across different experimental settings. Detailed results are provided in the full version.

*1) Conflict Metrics:* We replace polarization with disagreement as the conflict objective. The overall performance pattern remains consistent with the main results, with *CRD* remaining competitive and *CRD*$^+$ achieving the lowest disagreement on all larger datasets.

*2) Diffusion Models:* We replace the IC model with the LT model to examine the effect of the diffusion process. The proposed methods retain similar relative performance, with *CRD*$^+$ achieving the lowest polarization on all larger datasets.

*3) Opinion Distributions:* We replace the default uniform innate opinions with fixed exponential-style opinion vectors.

TABLE III: Ablation Study of *CRD* on Final Polarization

| Dataset | *CRD* | *CRD′* | *CRD″* | *CRD*$^{\dagger}$ |
|---|---|---|---|---|
| *Convote* | 2.26 | 2.24 | 2.24 | 2.26 |
| *NetScience* | 4.90 | 4.90 | 4.88 | 4.91 |
| *WikiTalkHT* | 3.71 | 3.79 | 3.69 | 3.74 |
| *WikiVote* | 7.00 | 7.10 | 6.92 | 7.19 |
| *Reed98* | 1.29 | 1.31 | 1.17 | 1.24 |
| *EmailUniv* | 5.84 | 5.99 | 5.73 | 5.82 |
| *Hamster* | 7.43 | 7.68 | 7.24 | 7.54 |
| *USFCA72* | 3.23 | 3.18 | 2.95 | 3.15 |
| *NIPS-Ego* | 33.26 | 33.66 | 33.26 | 33.80 |

The overall performance remains broadly consistent, indicating that the effectiveness of the proposed methods is robust to different innate-opinion distributions.

### *F. Ablation Study*

To examine the contribution of the main components of *CRD*, we construct three variants that modify the attack-pool update, defense search, and mixed-attack distribution, respectively:

- *CRD′*: A variant that retains only the most recently generated attack instead of the cumulative attack pool.
- *CRD″*: A variant that removes the empirical defense search and selects the defense solely using the surrogate-based branch.
- *CRD*$^{\dagger}$: A variant that replaces the uniform distribution over the attack pool with an adaptive mixed-attack distribution. The detailed construction is provided in the full version.

Table III reports the final polarization of *CRD* and its ablation variants. Compared with *CRD′*, *CRD* achieves lower polarization on most datasets, suggesting the benefit of maintaining the cumulative attack pool. *CRD″* yields comparable or lower polarization across the evaluated datasets, suggesting that the empirical defense search does not consistently improve point-estimate solution quality. The adaptive mixed-attack variant *CRD*$^{\dagger}$ provides no consistent improvement in polarization. *CRD′* and *CRD″* generally require less running time than *CRD*, whereas *CRD*$^{\dagger}$ is substantially slower. Detailed results on running time are provided in the full version. Overall, the results reveal different trade-offs between effectiveness and computational cost among the components of *CRD*.

### *G. Effects of FJ Computation and Linearization*

To separately examine the effects of FJ computation and objective linearization, we evaluate both *CRD* and *CRD*$^{+}$ using direct and iterative FJ computation on the same datasets. This comparison isolates the impact of the FJ computation method from that of the linearization used in *CRD*$^{+}$.

Table IV reports the final polarization. Direct and iterative FJ computation yield similar polarization for both methods, while *CRD*$^{+}$ generally achieves lower polarization than *CRD*.

Table V reports the corresponding running times. Direct FJ computation is generally faster for *CRD*, whereas the two approaches have comparable running times for *CRD*$^{+}$. Moreover, *CRD*$^{+}$ requires substantially less running time than *CRD* across all evaluated datasets. Overall, the FJ computation method has limited impact on solution quality, while the linearization in *CRD*$^{+}$ substantially reduces computational cost without degrading solution quality in these experiments.

TABLE IV: Final Polarization under Different FJ Computation Methods

| Dataset | Method | Direct FJ | Iterative FJ |
|---|---|---|---|
| *Convote* | *CRD* | $2.26 \pm 0.03$ | $2.25 \pm 0.04$ |
| | *CRD*$^{+}$ | $2.23 \pm 0.01$ | $2.21 \pm 0.02$ |
| *NetScience* | *CRD* | $4.90 \pm 0.07$ | $4.96 \pm 0.08$ |
| | *CRD*$^{+}$ | $4.86 \pm 0.01$ | $4.93 \pm 0.02$ |
| *WikiTalkHT* | *CRD* | $3.71 \pm 0.04$ | $3.71 \pm 0.16$ |
| | *CRD*$^{+}$ | $3.70 \pm 0.02$ | $3.59 \pm 0.05$ |
| *WikiVote* | *CRD* | $7.00 \pm 0.24$ | $6.99 \pm 0.28$ |
| | *CRD*$^{+}$ | $6.92 \pm 0.01$ | $6.91 \pm 0.03$ |
| *Reed98* | *CRD* | $1.29 \pm 0.10$ | $1.29 \pm 0.13$ |
| | *CRD*$^{+}$ | $1.16 \pm 0.00$ | $1.20 \pm 0.01$ |
| *EmailUniv* | *CRD* | $5.84 \pm 0.21$ | $6.07 \pm 0.29$ |
| | *CRD*$^{+}$ | $5.72 \pm 0.03$ | $5.72 \pm 0.03$ |

TABLE V: Running Time under Different FJ Computation Methods (Seconds)

| Dataset | Method | Direct FJ | Iterative FJ |
|---|---|---|---|
| *Convote* | *CRD* | 82.44 | 210.03 |
| | *CRD*$^{+}$ | 3.47 | 3.08 |
| *NetScience* | *CRD* | 163.28 | 341.77 |
| | *CRD*$^{+}$ | 2.21 | 2.08 |
| *WikiTalkHT* | *CRD* | 231.52 | 710.19 |
| | *CRD*$^{+}$ | 6.04 | 5.53 |
| *WikiVote* | *CRD* | 1303.84 | 3149.94 |
| | *CRD*$^{+}$ | 5.71 | 6.07 |
| *Reed98* | *CRD* | 1854.50 | 5670.26 |
| | *CRD*$^{+}$ | 65.78 | 56.26 |
| *EmailUniv* | *CRD* | 4474.13 | 6464.75 |
| | *CRD*$^{+}$ | 16.24 | 12.78 |

## VI. Conclusion

We studied robust counter-cascade defense against adaptive opinion-conflict attacks in online social networks. We formulated the RCD problem under a Stackelberg counter-cascade model that combines stochastic propagation with FJ opinion dynamics. We developed *CRD*, which constructs monotone submodular surrogates from a realization-wise decomposition of the quadratic conflict objective and employs weighted RIS within a constraint-generation framework. Its lower and upper certificates provide probabilistic guarantees, certifying an $\varepsilon$-near-optimal defense when the certificate gap reaches the prescribed tolerance. We further introduced *CRD*$^{+}$ as a practical variant for improved scalability. Extensive experiments on real-world networks demonstrate the strong defense effectiveness, robustness, and scalability of the proposed methods.

Future work could explore complementary extensions of our framework to richer cascade interactions, including blocking diffusion and temporal propagation, as well as opinion-dynamics models beyond FJ. Another direction is to develop tighter surrogates and more sample-efficient certified optimization methods to improve the scalability of *CRD* on large networks.

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The authors used ChatGPT (OpenAI) to assist with code development, technical refinement, and language editing. The experimental code underlying Section V was developed through iterative human–AI interaction, with the authors reviewing its overall architecture and critical implementation details. The authors developed the overall algorithmic frameworks and proof strategies, while ChatGPT assisted in refining their technical details and presentation in Section IV and the associated proofs in the full version. ChatGPT was also used for grammar correction and language polishing throughout the manuscript. The authors reviewed and verified the AI-assisted code, algorithms, and proofs and take full responsibility for the correctness, originality, and integrity of the work and its reported results.

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## APPENDIX A
## ADAPTIVE MIXED-ATTACK DISTRIBUTIONS

The main text uses the uniform distribution over the attack pool $\mathcal{A}$ as the default mixed attack distribution. This choice is simple and already gives a valid lower certificate. The certificate validity, however, does not require $\pi$ to be uniform. For any distribution $\pi$ over $\mathcal{A}$, Lemma 9 gives a valid lower certificate $L_{\mathrm{mix}}(\pi)$. Therefore, the defense oracle may generate several distributions and keep the largest valid lower certificate.

A practical adaptive strategy is to increase the mass on attacks that remain hard for the current mixed surrogate defense. Let $\mathcal{A} = \{S_A^1, \dots, S_A^p\}$. Initialize $\pi_i^1 = 1/p$ for every $i \in [p]$. At round $r$, run Algorithm 2 with distribution $\pi^r$, obtaining $S_B^\circ(\pi^r)$ and $L_{\mathrm{mix}}(\pi^r)$. For each attack $S_A^i$, compute a score $\chi_i^r = h_i(S_B^\circ(\pi^r))$, which measures the fixed-attack lower payoff under the current surrogate defense. Normalize the scores into $[0,1]$, obtaining $\tilde{\chi}_i^r$. Then update

$$\pi_i^{r+1} = \frac{\pi_i^r \exp(\gamma_\pi \tilde{\chi}_i^r)}{\sum_{j=1}^p \pi_j^r \exp(\gamma_\pi \tilde{\chi}_j^r)},$$

where $\gamma_\pi > 0$ is a learning-rate parameter.

This update gives more weight to attacks with larger lower payoffs under the current surrogate defense. After a fixed number of rounds, the defense oracle may use the best certificate

$$L = \max_r L_{\mathrm{mix}}(\pi^r).$$

This maximum is still a valid lower certificate because each $L_{\mathrm{mix}}(\pi^r)$ is individually valid. When sample-based estimates are used, the failure probability is allocated across all adaptive rounds so that every retained certificate is valid with the prescribed overall probability.

## APPENDIX B
## MISSING PROOFS

This section provides the omitted proofs of the theoretical results presented in the main text.

*Proof of Lemma 1.* Recall that $Q = (I + L)^{-1}$. Since $L \succeq 0$, all eigenvalues of $Q$ lie in $(0, 1]$, and hence $0 \preceq Q \preceq I$.

For polarization, $P = Q(I - \mathbf{1}\mathbf{1}^\top/n)Q$. The matrix $I - \mathbf{1}\mathbf{1}^\top/n$ is an orthogonal projection, so $0 \preceq I - \mathbf{1}\mathbf{1}^\top/n \preceq I$. Therefore, $0 \preceq P \preceq Q^2 \preceq I$.

For disagreement, $D = QLQ$. Since $L = Q^{-1} - I$, we have $D = Q(Q^{-1} - I)Q = Q - Q^2$. If $\lambda \geq 0$ is an eigenvalue of $L$, then the corresponding eigenvalue of $D$ is $\lambda/(1+\lambda)^2$, which lies in $[0, 1]$. Thus, $0 \preceq D \preceq I$. □

TABLE VI: Running Time under the Default Setting (Seconds)

| Dataset | *CRD/CRD*$^+$ | *FJ-Iter* | *HIST-Iter* |
|---|---|---|---|
| *Convote* | 82.44 | 7.62 | 0.34 |
| *NetScience* | 163.28 | 5.91 | 0.41 |
| *WikiTalkHT* | 231.52 | 14.48 | 0.22 |
| *WikiVote* | 1303.84 | 26.46 | 0.48 |
| *Reed98* | 1854.50 | 101.04 | 4.09 |
| *EmailUniv* | 4474.13 | 53.95 | 1.16 |
| *Hamster* | 16541.74 | 152.19 | 2.17 |
| *USFCA72* | 36135.65 | 478.42 | 9.61 |
| *NIPS-Ego* | 29764.71 | 84.27 | 0.94 |
| *PagesGov* | 154.03 | — | 7.85 |
| *HepPh* | 177.59 | — | 9.06 |
| *AnyBeat* | 303.45 | — | 9.13 |
| *CondMat* | 151.50 | — | 7.72 |
| *GPlus* | 144.49 | — | 6.08 |
| *Brightkite* | 309.35 | — | 27.36 |
| *WikiTalk* | 607.23 | — | 58.66 |
| *Epinions* | 657.56 | — | 104.30 |
| *Gowalla* | 1822.97 | — | 230.13 |
| *YouTube* | 19196.08 | — | 5696.13 |
| *Pokec* | 55911.03 | — | 16139.21 |
| *Orkut* | 252467.58 | — | 70375.80 |

*Proof of Lemma 2.* Fix a realization $g$. By Lemma 1, we have $0 \preceq C \preceq I$. Therefore,

$$0 \leq (\xi^A \odot r^A)^\top C (\xi^A \odot r^A) \leq (\xi^A \odot r^A)^\top (\xi^A \odot r^A).$$

Since $r_u^A \in \{0, 1\}$, this implies

$$\sum_{u,v \in V} M_{uv}^A r_u^A r_v^A \leq \sum_{u \in V} \left(\xi_u^A\right)^2 r_u^A.$$

Combining this inequality with Eq. (5), we obtain

$$\hat{s}^\top C \hat{s} \leq (\eta^A)^\top C \eta^A + \sum_{u \in V} \left(\ell_u^A + \left(\xi_u^A\right)^2\right) r_u^A.$$

Since $r_u^A \geq 0$, replacing each coefficient $\ell_u^A + (\xi_u^A)^2$ by its positive part can only increase the summation. Thus, by the definition of $\psi_u^A$,

$$\hat{s}^\top C \hat{s} \leq (\eta^A)^\top C \eta^A + \sum_{u \in V} \psi_u^A r_u^A.$$

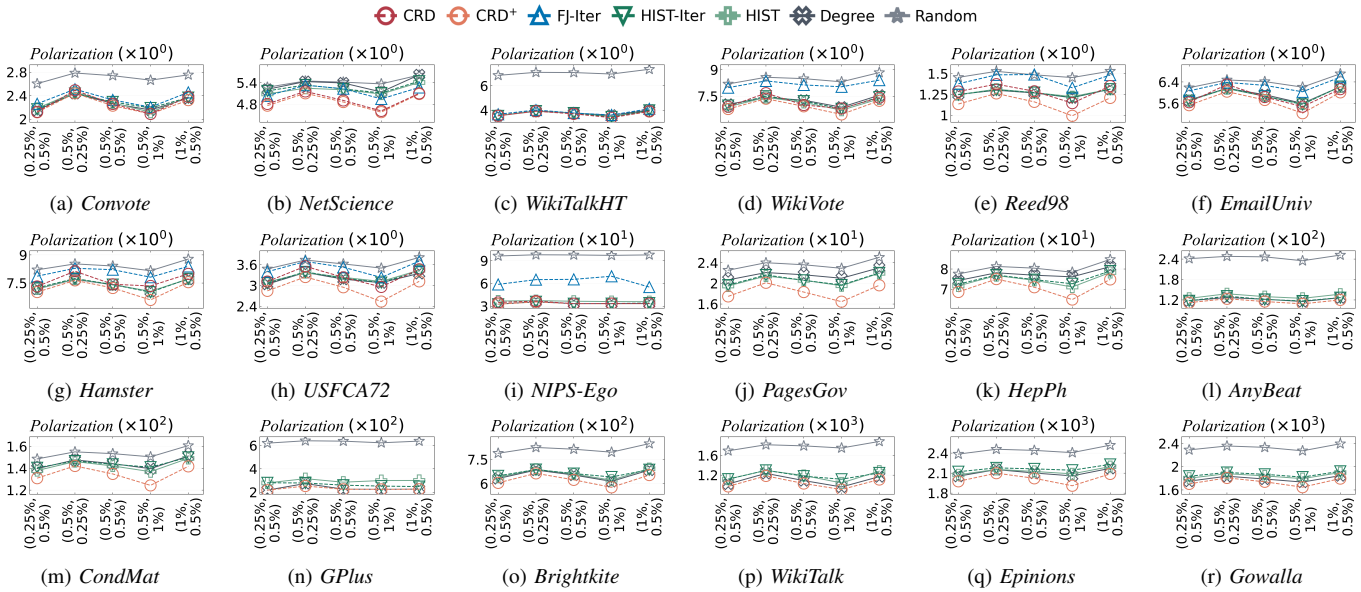
Taking expectation over $g \sim \mathcal{G}$ gives Eq. (7). □

*Proof of Lemma 3.* For fixed $S_B$ and realization $g$, the weights $\psi_u^A$ are fixed nonnegative quantities by definition and are independent of the choice of $S_A$. For every realization $g$, the function $\sum_{u \in V} \psi_u^A r_u^A$ is a nonnegative weighted coverage function over the nodes reached by $S_A$, and is therefore monotone submodular in $S_A$. Taking a nonnegative expectation preserves monotonicity and submodularity. □

*Proof of Lemma 4.* It suffices to consider a single weighted RR-set sample. Conditioned on a realization $g_i$, the root node $u_i$ is sampled uniformly from $V$. Therefore,

$$\begin{aligned}&\mathbb{E}_{u_i}\left[n\psi_{u_i}^A \mathbf{1}[R_{u_i}^A(g_i) \cap S_A \neq \emptyset] \mid g_i\right] \\ &= \sum_{u \in V} \psi_u^A \mathbf{1}[R_u^A(g_i) \cap S_A \neq \emptyset].\end{aligned}$$

By definition, $u$ is reachable from $S_A$ in $g_i$ if and only if $R_u^A(g_i) \cap S_A \neq \emptyset$. Hence, $\mathbf{1}[R_u^A(g_i) \cap S_A \neq \emptyset] = r_u^A$. Taking

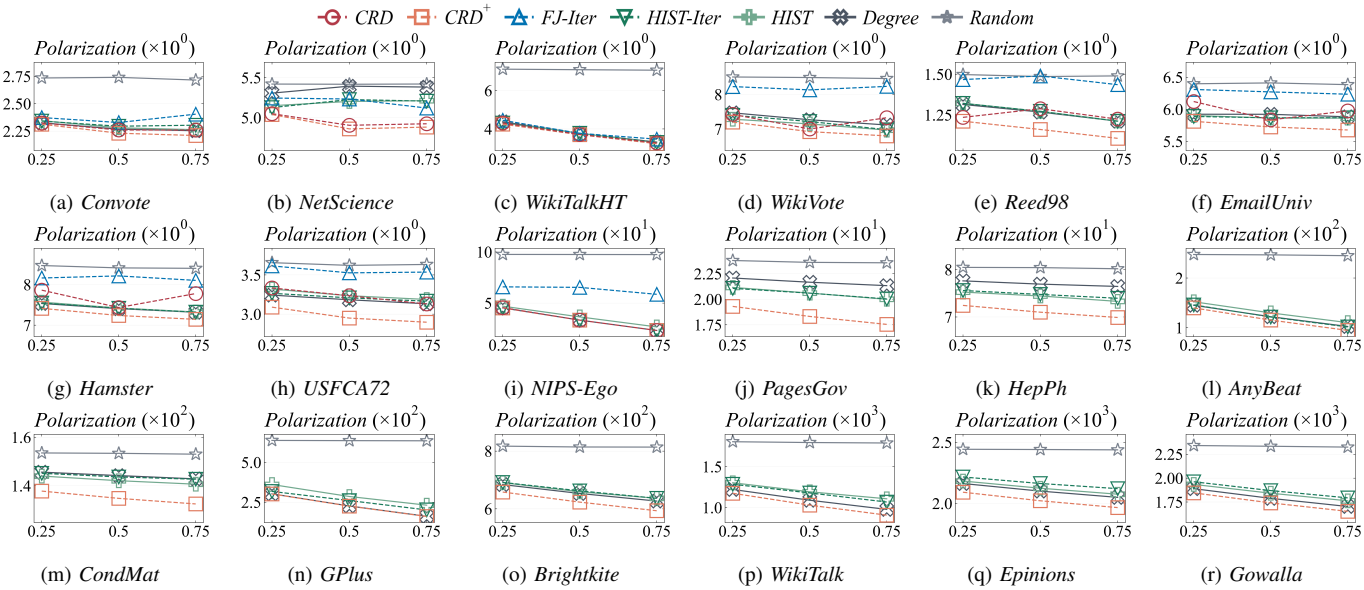



(a) *Convote* (b) *NetScience* (c) *WikiTalkHT* (d) *WikiVote* (e) *Reed98* (f) *EmailUniv*

(g) *Hamster* (h) *USFCA72* (i) *NIPS-Ego* (j) *PagesGov* (k) *HepPh* (l) *AnyBeat*

(m) *CondMat* (n) *GPlus* (o) *Brightkite* (p) *WikiTalk* (q) *Epinions* (r) *Gowalla*

Fig. 4: Final polarization vs. asymmetric seed budgets $(k_A, k_B)$, expressed as percentages of $n$.




(a) *Convote* (b) *NetScience* (c) *WikiTalkHT* (d) *WikiVote* (e) *Reed98* (f) *EmailUniv*

(g) *Hamster* (h) *USFCA72* (i) *NIPS-Ego* (j) *PagesGov* (k) *HepPh* (l) *AnyBeat*

(m) *CondMat* (n) *GPlus* (o) *Brightkite* (p) *WikiTalk* (q) *Epinions* (r) *Gowalla*

Fig. 5: Final polarization vs. $\rho$ for different methods.

expectation over $g_i \sim \mathcal{G}$ and using Eq. (6) proves the claim for one sample. The lemma follows by linearity of expectation over the $\theta_A$ independent samples. □

*Proof of Lemma 5.* Since $C \in \{P, D\}$ has zero row sums and $\eta^A \in [0,1]^n$, the definition of $\beta$ gives $|(C\eta^A)_u| \le \beta$ for every $u \in V$. Moreover, $|\xi_u^A| \le 1$, and hence $|\ell_u^A| \le 2\beta$. Therefore, $\psi_u^A \le 2\beta + 1$, so each single-sample term in Eq. (9) lies in $[0, n(2\beta+1)]$.

Let $S_A^\star \in \arg\max_{S_A \in \mathcal{S}_A} \Psi^A(S_A \mid S_B)$. For the fixed set $S_A^\star$, Hoeffding's inequality [74] gives

$$\Pr\Big[\Psi^A(S_A^\star \mid S_B) > \widehat{\Psi}^A(S_A^\star \mid S_B) + \varepsilon_A\Big] \le \exp\left(-\frac{2\theta_A \varepsilon_A^2}{n^2(2\beta+1)^2}\right).$$

By the sample size condition in Eq. (11), the right-hand side is at most $\delta_1$. Thus, with probability at least $1 - \delta_1$,

$$\Psi^A(S_A^\star \mid S_B) \le \widehat{\Psi}^A(S_A^\star \mid S_B) + \varepsilon_A. \tag{19}$$

For every realization of the sampled RR-sets, the empirical objective in Eq. (9) is a monotone submodular weighted cov-

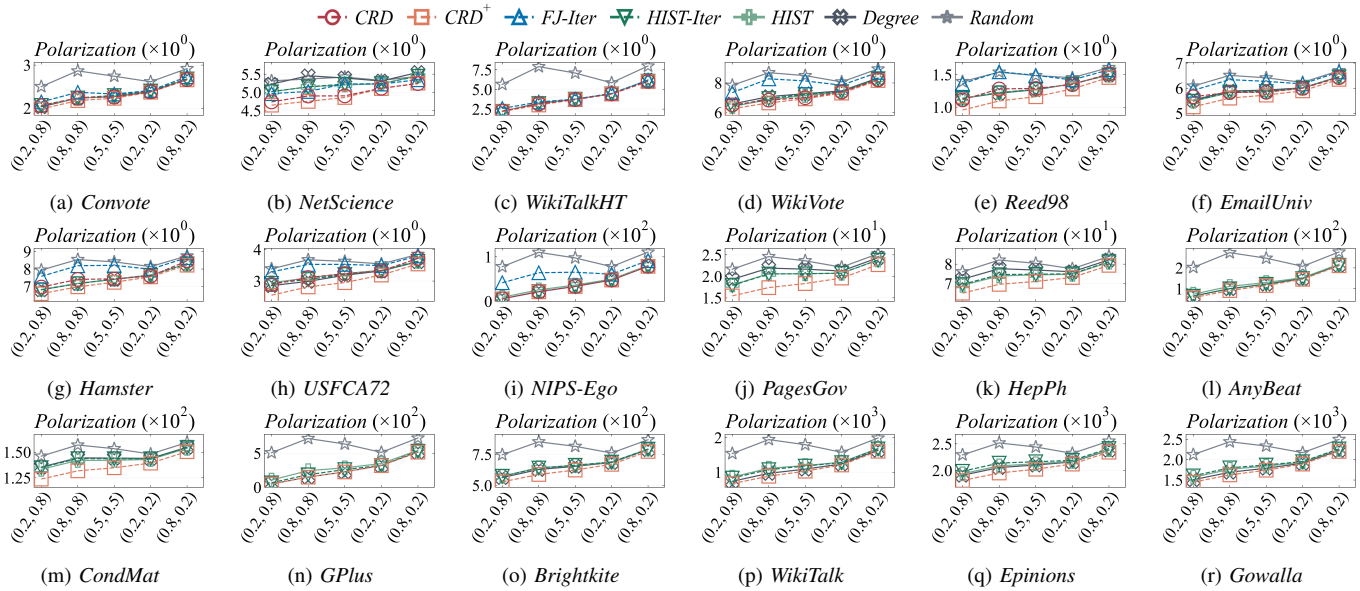



Fig. 6: Final polarization vs. $(\alpha_A, \alpha_B)$ for different methods.

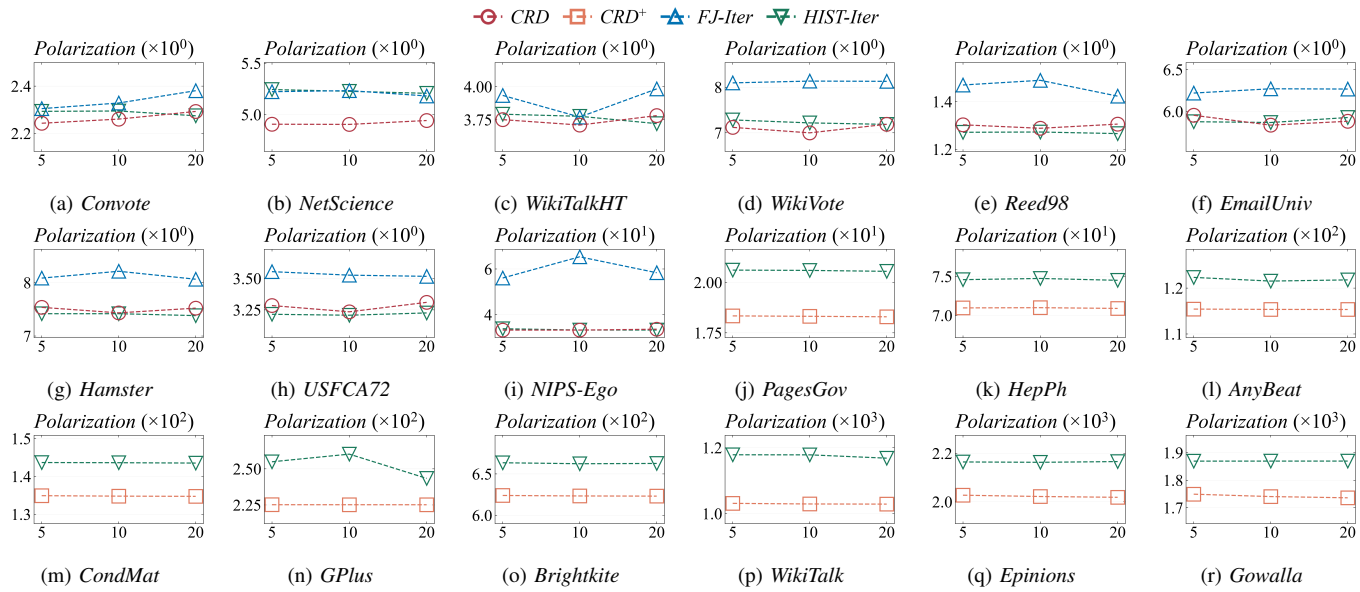



Fig. 7: Final polarization vs. outer iteration limit $T$ for different methods.

erage function. Therefore, the greedy selection in Algorithm 1 satisfies

$$\widehat{\Psi}^A(S_A^\circ \mid S_B) \geq \left(1 - \frac{1}{e}\right) \widehat{\Psi}^A(S_A^\star \mid S_B). \tag{20}$$

Combining Eq. (19) and Eq. (20), we obtain

$$\begin{aligned} \Psi^A_{\max}(S_B) &= \Psi^A(S_A^\star \mid S_B) \\ &\leq \widehat{\Psi}^A(S_A^\star \mid S_B) + \varepsilon_A \\ &\leq \frac{\widehat{\Psi}^A(S_A^\circ \mid S_B)}{1 - \frac{1}{e}} + \varepsilon_A. \end{aligned} \tag{21}$$

Combining Eq. (21) with Eq. (8) and Eq. (10) gives

$$\Phi(S_B) \leq K^A(S_B) + \Psi^A_{\max}(S_B) \leq U(S_B),$$

which proves the claim. □

*Proof of Lemma 6.* Fix a realization $g$. By Lemma 1, we have $C \succeq 0$. Therefore,

$$\sum_{u,v \in V} M^B_{uv} r^B_u r^B_v = (\xi^B \odot r^B)^\top C (\xi^B \odot r^B) \geq 0.$$

Combining this inequality with Eq. (5), we obtain

$$\hat{s}^\top C \hat{s} \geq (\eta^B)^\top C \eta^B + \sum_{u \in V} \ell^B_u r^B_u.$$

TABLE VII: Final Polarization under Different Sampling Budgets

| Dataset | Method | 1,000 | 10,000 | 100,000 | 1,000,000 |
|---|---|---|---|---|---|
| *Convote* | *CRD* | $2.40 \pm 0.20$ | $2.30 \pm 0.12$ | $2.26 \pm 0.03$ | $2.23 \pm 0.01$ |
| *NetScience* | *CRD* | $5.18 \pm 0.21$ | $5.18 \pm 0.25$ | $4.90 \pm 0.07$ | $4.86 \pm 0.01$ |
| *WikiTalkHT* | *CRD* | $5.56 \pm 1.26$ | $3.81 \pm 0.23$ | $3.71 \pm 0.04$ | $3.69 \pm 0.02$ |
| *WikiVote* | *CRD* | $8.08 \pm 0.45$ | $7.63 \pm 0.75$ | $7.00 \pm 0.24$ | $6.95 \pm 0.08$ |
| *Reed98* | *CRD* | $1.34 \pm 0.09$ | $1.34 \pm 0.03$ | $1.29 \pm 0.10$ | $1.18 \pm 0.04$ |
| *EmailUniv* | *CRD* | $6.06 \pm 0.15$ | $6.03 \pm 0.27$ | $5.84 \pm 0.21$ | $5.79 \pm 0.11$ |
| *Hamster* | *CRD* | $7.86 \pm 0.29$ | $7.87 \pm 0.54$ | $7.43 \pm 0.42$ | $7.47 \pm 0.39$ |
| *USFCA72* | *CRD* | $3.35 \pm 0.16$ | $3.34 \pm 0.04$ | $3.23 \pm 0.21$ | $3.08 \pm 0.24$ |
| *NIPS-Ego* | *CRD* | $78.17 \pm 13.45$ | $64.22 \pm 22.93$ | $33.26 \pm 0.00$ | $33.31 \pm 0.16$ |
| *PagesGov* | *CRD*$^+$ | $18.87 \pm 0.30$ | $18.56 \pm 0.17$ | $18.32 \pm 0.05$ | $18.28 \pm 0.01$ |
| *HepPh* | *CRD*$^+$ | $71.71 \pm 0.40$ | $70.96 \pm 0.22$ | $70.99 \pm 0.16$ | $70.94 \pm 0.08$ |
| *AnyBeat* | *CRD*$^+$ | $115.91 \pm 0.48$ | $115.78 \pm 0.42$ | $115.34 \pm 0.21$ | $115.15 \pm 0.08$ |
| *CondMat* | *CRD*$^+$ | $137.05 \pm 0.79$ | $135.28 \pm 0.44$ | $134.79 \pm 0.09$ | $134.70 \pm 0.14$ |
| *GPlus* | *CRD*$^+$ | $228.45 \pm 2.17$ | $225.59 \pm 0.38$ | $225.16 \pm 0.04$ | $225.12 \pm 0.01$ |
| *Brightkite* | *CRD*$^+$ | $621.66 \pm 1.58$ | $621.35 \pm 1.20$ | $623.32 \pm 0.52$ | $621.67 \pm 0.29$ |
| *WikiTalk* | *CRD*$^+$ | $1019.92 \pm 3.61$ | $1026.71 \pm 1.64$ | $1029.21 \pm 1.04$ | $1025.01 \pm 0.94$ |
| *Epinions* | *CRD*$^+$ | $2024.70 \pm 2.34$ | $2029.32 \pm 2.59$ | $2021.71 \pm 1.40$ | $2020.74 \pm 0.80$ |
| *Gowalla* | *CRD*$^+$ | $1748.98 \pm 3.47$ | $1744.09 \pm 1.80$ | $1740.48 \pm 0.94$ | $1739.28 \pm 0.75$ |

TABLE VIII: Running Time under Different Sampling Budgets (Seconds)

| Dataset | Method | 1,000 | 10,000 | 100,000 | 1,000,000 |
|---|---|---|---|---|---|
| *Convote* | *CRD* | 0.96 | 8.26 | 82.44 | 790.14 |
| *NetScience* | *CRD* | 1.64 | 16.97 | 163.28 | 1467.55 |
| *WikiTalkHT* | *CRD* | 2.46 | 24.27 | 231.52 | 2256.23 |
| *WikiVote* | *CRD* | 11.98 | 121.97 | 1303.84 | 10541.81 |
| *Reed98* | *CRD* | 22.30 | 205.88 | 1854.50 | 12409.25 |
| *EmailUniv* | *CRD* | 22.67 | 297.92 | 4474.13 | 21615.23 |
| *Hamster* | *CRD* | 117.40 | 1263.45 | 16541.74 | 150318.13 |
| *USFCA72* | *CRD* | 265.73 | 2907.28 | 36135.65 | 359275.04 |
| *NIPS-Ego* | *CRD* | 320.40 | 3394.03 | 29764.71 | 290245.87 |
| *PagesGov* | *CRD*$^+$ | 7.14 | 24.10 | 154.03 | 1985.88 |
| *HepPh* | *CRD*$^+$ | 16.93 | 38.06 | 177.59 | 2184.72 |
| *AnyBeat* | *CRD*$^+$ | 6.91 | 37.21 | 303.45 | 3227.94 |
| *CondMat* | *CRD*$^+$ | 11.87 | 27.10 | 151.50 | 1279.94 |
| *GPlus* | *CRD*$^+$ | 3.98 | 23.89 | 144.49 | 554.15 |
| *Brightkite* | *CRD*$^+$ | 44.36 | 113.55 | 309.35 | 5234.88 |
| *WikiTalk* | *CRD*$^+$ | 74.70 | 205.28 | 607.23 | 9434.57 |
| *Epinions* | *CRD*$^+$ | 130.01 | 428.45 | 657.56 | 7560.14 |
| *Gowalla* | *CRD*$^+$ | 228.56 | 817.34 | 1822.97 | 16494.15 |

TABLE IX: Final Polarization under Different Certificate Tolerances

| Dataset | $\varepsilon/n$ | | |
|---|---|---|---|
| | 2% | 5% | 10% |
| *Convote* | 2.23 | 2.24 | 2.24 |
| *NetScience* | 4.87 | 4.88 | 4.86 |
| *WikiTalkHT* | 3.69 | 3.77 | 3.76 |
| *WikiVote* | 6.94 | 6.94 | 7.01 |
| *Reed98* | 1.17 | 1.17 | 1.21 |
| *EmailUniv* | 5.74 | 5.74 | 5.83 |

TABLE X: Running Time under Different Certificate Tolerances (Seconds)

| Dataset | $\varepsilon/n$ | | |
|---|---|---|---|
| | 2% | 5% | 10% |
| *Convote* | 5448.13 | 72.78 | 18.27 |
| *NetScience* | 1291.09 | 205.57 | 51.51 |
| *WikiTalkHT* | 34009.26 | 401.11 | 100.00 |
| *WikiVote* | 14807.77 | 2397.58 | 609.00 |
| *Reed98* | 8815.61 | 1440.84 | 351.65 |
| *EmailUniv* | 21191.51 | 3771.75 | 876.25 |

For each $u$, since $r_u^B \geq 0$, we have $\ell_u^B r_u^B \geq -[-\ell_u^B]_+ r_u^B$. By the definition of $\psi_u^B$, this implies $\ell_u^B r_u^B \geq -\psi_u^B r_u^B$. Thus,

$$\hat{s}^\top C \hat{s} \geq (\eta^B)^\top C \eta^B - \sum_{u \in V} \psi_u^B r_u^B.$$

Taking expectation over $g \sim \mathcal{G}$ gives

$$F(S_A, S_B) \geq K^B(S_A) - \Psi^B(S_B \mid S_A).$$

By Eq. (13), this proves Eq. (14). □

*Proof of Lemma 7.* For fixed $S_A$ and realization $g$, the weights $\psi_u^B$ are fixed nonnegative quantities by definition and are independent of the choice of $S_B$. For every realization $g$, the function $\sum_{u \in V} \psi_u^B r_u^B$ is a nonnegative weighted coverage function over the nodes reached by $S_B$, and is therefore monotone submodular in $S_B$. Taking a nonnegative expectation preserves monotonicity and submodularity. □

*Proof of Lemma 8.* It suffices to consider a single mixed weighted RR-set sample. Conditioned on the sampled attack $T_q = S_A^i$ and realization $g_q$, the root node $u_q$ is sampled

TABLE XI: Final Disagreement under the Default Setting

| Dataset | *CRD/CRD$^+$* | *FJ-Iter* | *HIST-Iter* | *HIST* | *Degree* | *Random* |
|---|---|---|---|---|---|---|
| *Convote* | $3.07 \pm 0.03$ | $3.14 \pm 0.07$ | $3.12 \pm 0.04$ | $3.11 \pm 0.04$ | $\mathbf{3.07 \pm 0.01}$ | $3.63 \pm 0.08$ |
| *NetScience* | $4.86 \pm 0.13$ | $5.07 \pm 0.13$ | $\mathbf{4.78 \pm 0.01}$ | $4.78 \pm 0.01$ | $5.06 \pm 0.00$ | $5.26 \pm 0.03$ |
| *WikiTalkHT* | $5.11 \pm 0.07$ | $5.17 \pm 0.12$ | $5.20 \pm 0.14$ | $5.15 \pm 0.09$ | $\mathbf{5.03 \pm 0.01}$ | $9.43 \pm 0.58$ |
| *WikiVote* | $10.76 \pm 0.55$ | $12.14 \pm 0.43$ | $10.83 \pm 0.15$ | $\mathbf{10.73 \pm 0.10}$ | $10.83 \pm 0.01$ | $12.89 \pm 0.09$ |
| *Reed98* | $4.31 \pm 0.18$ | $4.79 \pm 0.13$ | $4.28 \pm 0.01$ | $4.31 \pm 0.03$ | $\mathbf{4.28 \pm 0.01}$ | $4.86 \pm 0.06$ |
| *EmailUniv* | $\mathbf{11.46 \pm 0.20}$ | $12.26 \pm 0.16$ | $11.51 \pm 0.04$ | $11.50 \pm 0.06$ | $11.51 \pm 0.04$ | $12.45 \pm 0.10$ |
| *Hamster* | $17.07 \pm 1.05$ | $18.35 \pm 0.34$ | $\mathbf{16.53 \pm 0.09}$ | $16.65 \pm 0.11$ | $16.57 \pm 0.02$ | $19.20 \pm 0.16$ |
| *USFCA72* | $11.10 \pm 0.70$ | $12.02 \pm 0.12$ | $10.80 \pm 0.10$ | $10.85 \pm 0.07$ | $\mathbf{10.65 \pm 0.03}$ | $12.14 \pm 0.10$ |
| *NIPS-Ego* | $\mathbf{33.39 \pm 0.01}$ | $65.63 \pm 16.07$ | $33.43 \pm 0.01$ | $36.20 \pm 2.09$ | $33.41 \pm 0.01$ | $97.59 \pm 0.10$ |
| *PagesGov* | $\mathbf{47.31 \pm 0.09}$ | — | $49.04 \pm 0.39$ | $49.21 \pm 0.33$ | $51.10 \pm 0.04$ | $57.20 \pm 0.23$ |
| *HepPh* | $\mathbf{116.71 \pm 0.15}$ | — | $120.53 \pm 0.26$ | $119.57 \pm 0.55$ | $124.49 \pm 0.07$ | $131.45 \pm 0.37$ |
| *AnyBeat* | $\mathbf{141.78 \pm 0.07}$ | — | $148.45 \pm 1.36$ | $158.82 \pm 2.82$ | $148.23 \pm 0.06$ | $292.38 \pm 1.46$ |
| *CondMat* | $\mathbf{235.77 \pm 0.34}$ | — | $244.92 \pm 0.65$ | $242.24 \pm 0.73$ | $243.78 \pm 0.18$ | $266.50 \pm 0.50$ |
| *GPlus* | $\mathbf{270.10 \pm 0.08}$ | — | $285.89 \pm 12.04$ | $334.27 \pm 13.32$ | $270.80 \pm 0.03$ | $728.01 \pm 7.55$ |
| *Brightkite* | $\mathbf{721.03 \pm 0.59}$ | — | $765.48 \pm 3.48$ | $766.05 \pm 4.06$ | $753.74 \pm 0.42$ | $956.81 \pm 2.52$ |
| *WikiTalk* | $\mathbf{1198.92 \pm 0.92}$ | — | $1350.13 \pm 7.19$ | $1374.88 \pm 30.17$ | $1259.23 \pm 1.03$ | $2067.95 \pm 7.49$ |
| *Epinions* | $\mathbf{1759.98 \pm 1.29}$ | — | $1911.01 \pm 6.36$ | $1858.15 \pm 5.12$ | $1816.66 \pm 0.60$ | $2299.33 \pm 5.22$ |
| *Gowalla* | $\mathbf{2278.79 \pm 0.88}$ | — | $2437.83 \pm 4.33$ | $2404.14 \pm 5.32$ | $2322.98 \pm 0.51$ | $3071.74 \pm 8.02$ |

TABLE XII: Final Polarization under the LT Model

| Dataset | *CRD/CRD$^+$* | *FJ-Iter* | *HIST-Iter* | *HIST* | *Degree* | *Random* |
|---|---|---|---|---|---|---|
| *Convote* | $2.28 \pm 0.03$ | $2.34 \pm 0.09$ | $2.29 \pm 0.04$ | $2.28 \pm 0.03$ | $\mathbf{2.26 \pm 0.01}$ | $2.78 \pm 0.09$ |
| *NetScience* | $\mathbf{4.88 \pm 0.09}$ | $5.06 \pm 0.13$ | $5.19 \pm 0.01$ | $5.18 \pm 0.02$ | $5.39 \pm 0.01$ | $5.43 \pm 0.06$ |
| *WikiTalkHT* | $\mathbf{3.55 \pm 0.01}$ | $3.69 \pm 0.12$ | $3.59 \pm 0.08$ | $3.62 \pm 0.12$ | $3.55 \pm 0.01$ | $6.73 \pm 1.09$ |
| *WikiVote* | $7.30 \pm 0.73$ | $8.13 \pm 0.40$ | $7.04 \pm 0.11$ | $\mathbf{7.00 \pm 0.09}$ | $7.07 \pm 0.02$ | $8.75 \pm 0.20$ |
| *Reed98* | $1.30 \pm 0.14$ | $1.55 \pm 0.12$ | $1.21 \pm 0.02$ | $1.21 \pm 0.03$ | $\mathbf{1.19 \pm 0.01}$ | $1.59 \pm 0.09$ |
| *EmailUniv* | $\mathbf{5.78 \pm 0.22}$ | $6.32 \pm 0.14$ | $5.84 \pm 0.08$ | $5.81 \pm 0.04$ | $5.81 \pm 0.02$ | $6.57 \pm 0.03$ |
| *Hamster* | $7.48 \pm 0.84$ | $8.43 \pm 0.21$ | $7.11 \pm 0.10$ | $7.18 \pm 0.09$ | $\mathbf{7.00 \pm 0.03}$ | $8.88 \pm 0.16$ |
| *USFCA72* | $3.29 \pm 0.46$ | $3.89 \pm 0.15$ | $2.88 \pm 0.05$ | $3.01 \pm 0.08$ | $\mathbf{2.75 \pm 0.03}$ | $4.06 \pm 0.09$ |
| *NIPS-Ego* | $\mathbf{33.17 \pm 0.00}$ | $52.38 \pm 7.69$ | $34.59 \pm 3.29$ | $37.59 \pm 2.21$ | $\mathbf{33.17 \pm 0.00}$ | $97.38 \pm 0.55$ |
| *PagesGov* | $\mathbf{17.18 \pm 0.03}$ | — | $19.13 \pm 0.34$ | $19.44 \pm 0.34$ | $20.13 \pm 0.06$ | $24.96 \pm 0.23$ |
| *HepPh* | $\mathbf{68.16 \pm 0.08}$ | — | $71.53 \pm 0.49$ | $72.37 \pm 0.39$ | $73.32 \pm 0.09$ | $83.39 \pm 0.54$ |
| *AnyBeat* | $\mathbf{107.87 \pm 0.19}$ | — | $112.38 \pm 0.81$ | $126.46 \pm 3.49$ | $113.43 \pm 0.11$ | $257.38 \pm 2.09$ |
| *CondMat* | $\mathbf{132.49 \pm 0.18}$ | — | $140.65 \pm 0.49$ | $141.25 \pm 0.87$ | $140.72 \pm 0.14$ | $156.47 \pm 0.45$ |
| *GPlus* | $\mathbf{219.38 \pm 0.01}$ | — | $261.09 \pm 61.35$ | $296.97 \pm 27.99$ | $220.30 \pm 0.01$ | $642.35 \pm 6.14$ |
| *Brightkite* | $\mathbf{592.22 \pm 0.40}$ | — | $627.92 \pm 1.92$ | $638.05 \pm 7.29$ | $625.90 \pm 0.81$ | $860.58 \pm 3.49$ |
| *WikiTalk* | $\mathbf{942.03 \pm 1.35}$ | — | $1061.34 \pm 7.89$ | $1099.66 \pm 5.88$ | $991.65 \pm 1.15$ | $1920.08 \pm 11.10$ |
| *Epinions* | $\mathbf{1976.76 \pm 1.07}$ | — | $2101.16 \pm 6.33$ | $2094.46 \pm 8.44$ | $2047.22 \pm 0.71$ | $2502.77 \pm 5.96$ |
| *Gowalla* | $\mathbf{1625.56 \pm 1.54}$ | — | $1753.67 \pm 5.32$ | $1737.36 \pm 8.81$ | $1673.27 \pm 1.12$ | $2467.30 \pm 37.64$ |

uniformly from $V$. Therefore,

$$\mathbb{E}_{u_q}\left[n\psi^B_{u_q}\mathbf{1}[R^B_{u_q}(g_q)\cap S_B\neq\emptyset] \mid T_q = S^i_A, g_q\right] = \sum_{u\in V}\psi^B_u\mathbf{1}[R^B_u(g_q)\cap S_B\neq\emptyset].$$

By the RR-set reachability equivalence, $\mathbf{1}[R^B_u(g_q)\cap S_B\neq\emptyset] = r^B_u$. Taking expectation over $g_q\sim\mathcal{G}$ gives $\Psi^B(S_B \mid S^i_A)$. Taking expectation over $T_q\sim\pi$ gives

$$\mathbb{E}\left[n\psi^B_{u_q}\mathbf{1}[R^B_{u_q}(g_q)\cap S_B\neq\emptyset]\right] = \sum_{i=1}^{p}\pi_i\Psi^B(S_B \mid S^i_A) = \Psi^B_\pi(S_B).$$

The result follows by linearity of expectation over the $\theta_B$ independent samples. □

*Proof of Lemma 9.* Since $C\in\{P,D\}$ has zero row sums and $\eta^B\in[0,1]^n$, the definition of $\beta$ gives $|(C\eta^B)_u|\le\beta$ for every $u\in V$. Moreover, $|\xi^B_u|\le 1$, and hence $|\ell^B_u|\le 2\beta$. Therefore, $\psi^B_u\le 2\beta$, so each single-sample term in Eq. (15) lies in $[0,2n\beta]$.

Let $S^\star_B\in\arg\max_{S_B\in\mathcal{S}_B}\Psi^B_\pi(S_B)$. For the fixed set $S^\star_B$, the one-sided Hoeffding inequality gives

$$\Pr\left[\Psi^B_\pi(S^\star_B) > \widehat{\Psi}^B_\pi(S^\star_B)+\varepsilon_B\right]\le\exp\left(-\frac{2\theta_B\varepsilon_B^2}{(2n\beta)^2}\right).$$

By the sample size condition in Eq. (17), the right-hand side is at most $\delta_2$. Thus, with probability at least $1-\delta_2$,

$$\Psi^B_\pi(S^\star_B)\le\widehat{\Psi}^B_\pi(S^\star_B)+\varepsilon_B. \tag{22}$$

For every realization of the sampled RR-sets, the empirical objective in Eq. (15) is a monotone submodular weighted coverage function. Therefore, the greedy selection in Algorithm 2 satisfies

$$\widehat{\Psi}^B_\pi(S^\circ_B(\pi))\ge\left(1-\frac{1}{e}\right)\widehat{\Psi}^B_\pi(S^\star_B). \tag{23}$$

Combining Eq. (22) and Eq. (23), we obtain

$$\begin{aligned}\Psi^B_{\max,\pi} &= \Psi^B_\pi(S^\star_B)\\ &\le\widehat{\Psi}^B_\pi(S^\star_B)+\varepsilon_B\\ &\le\frac{\widehat{\Psi}^B_\pi(S^\circ_B(\pi))}{1-\frac{1}{e}}+\varepsilon_B.\end{aligned} \tag{24}$$

TABLE XIII: Final Polarization with Exponential Innate Opinions

| Dataset | *CRD*/*CRD*$^+$ | *FJ-Iter* | *HIST-Iter* | *HIST* | *Degree* | *Random* |
|---|---|---|---|---|---|---|
| *Convote* | $\mathbf{1.24 \pm 0.03}$ | $1.27 \pm 0.03$ | $1.29 \pm 0.03$ | $1.27 \pm 0.04$ | $1.28 \pm 0.00$ | $1.44 \pm 0.03$ |
| *NetScience* | $1.30 \pm 0.06$ | $\mathbf{1.26 \pm 0.02}$ | $1.43 \pm 0.04$ | $1.46 \pm 0.03$ | $1.35 \pm 0.01$ | $1.29 \pm 0.03$ |
| *WikiTalkHT* | $\mathbf{1.26 \pm 0.06}$ | $1.82 \pm 0.48$ | $1.32 \pm 0.15$ | $1.34 \pm 0.14$ | $1.56 \pm 0.02$ | $2.19 \pm 0.08$ |
| *WikiVote* | $1.73 \pm 0.05$ | $1.80 \pm 0.06$ | $1.71 \pm 0.02$ | $\mathbf{1.70 \pm 0.02}$ | $1.76 \pm 0.01$ | $1.77 \pm 0.02$ |
| *Reed98* | $0.21 \pm 0.01$ | $0.22 \pm 0.01$ | $0.17 \pm 0.01$ | $0.18 \pm 0.01$ | $\mathbf{0.17 \pm 0.00}$ | $0.22 \pm 0.01$ |
| *EmailUniv* | $\mathbf{1.27 \pm 0.03}$ | $1.28 \pm 0.02$ | $1.28 \pm 0.01$ | $1.28 \pm 0.02$ | $1.29 \pm 0.01$ | $1.31 \pm 0.01$ |
| *Hamster* | $1.37 \pm 0.10$ | $1.43 \pm 0.07$ | $1.31 \pm 0.04$ | $1.29 \pm 0.02$ | $1.29 \pm 0.02$ | $\mathbf{1.28 \pm 0.04}$ |
| *USFCA72* | $0.80 \pm 0.05$ | $0.89 \pm 0.03$ | $0.81 \pm 0.02$ | $0.81 \pm 0.01$ | $\mathbf{0.79 \pm 0.01}$ | $0.92 \pm 0.01$ |
| *NIPS-Ego* | $\mathbf{4.39 \pm 0.00}$ | $16.56 \pm 2.89$ | $4.39 \pm 0.00$ | $8.32 \pm 1.99$ | $4.39 \pm 0.00$ | $14.68 \pm 1.10$ |
| *PagesGov* | $\mathbf{3.10 \pm 0.05}$ | — | $3.65 \pm 0.13$ | $3.68 \pm 0.17$ | $4.48 \pm 0.06$ | $3.75 \pm 0.05$ |
| *HepPh* | $\mathbf{11.59 \pm 0.19}$ | — | $14.63 \pm 0.25$ | $14.42 \pm 0.16$ | $15.01 \pm 0.05$ | $14.29 \pm 0.18$ |
| *AnyBeat* | $\mathbf{13.72 \pm 0.20}$ | — | $17.70 \pm 1.09$ | $20.83 \pm 1.73$ | $18.54 \pm 0.18$ | $23.72 \pm 0.54$ |
| *CondMat* | $\mathbf{16.21 \pm 0.19}$ | — | $19.58 \pm 0.44$ | $20.35 \pm 0.42$ | $20.37 \pm 0.18$ | $17.28 \pm 0.18$ |
| *GPlus* | $\mathbf{15.51 \pm 0.02}$ | — | $43.77 \pm 12.62$ | $61.95 \pm 4.65$ | $16.41 \pm 0.02$ | $37.68 \pm 8.00$ |
| *Brightkite* | $\mathbf{86.13 \pm 0.93}$ | — | $100.16 \pm 1.61$ | $99.64 \pm 1.11$ | $100.04 \pm 0.49$ | $112.31 \pm 0.43$ |
| *WikiTalk* | $\mathbf{144.51 \pm 2.27}$ | — | $205.41 \pm 2.51$ | $214.18 \pm 7.00$ | $181.34 \pm 1.24$ | $231.53 \pm 2.08$ |
| *Epinions* | $\mathbf{206.31 \pm 1.11}$ | — | $208.54 \pm 1.27$ | $216.46 \pm 1.45$ | $223.75 \pm 0.48$ | $218.38 \pm 0.66$ |
| *Gowalla* | $\mathbf{235.67 \pm 1.30}$ | — | $253.84 \pm 0.90$ | $251.44 \pm 1.08$ | $256.18 \pm 1.31$ | $290.06 \pm 0.87$ |

TABLE XIV: Running Time of *CRD* and Its Ablation Variants (Seconds)

| Dataset | *CRD* | *CRD′* | *CRD″* | *CRD*$^\dagger$ |
|---|---|---|---|---|
| *Convote* | 82.44 | 52.98 | 31.37 | 305.85 |
| *NetScience* | 163.28 | 95.41 | 64.67 | 684.63 |
| *WikiTalkHT* | 231.52 | 162.50 | 104.14 | 1015.09 |
| *WikiVote* | 1303.84 | 557.53 | 382.44 | 5238.07 |
| *Reed98* | 1854.50 | 1039.35 | 684.63 | 8174.10 |
| *EmailUniv* | 4474.13 | 1637.48 | 1748.51 | 24542.71 |
| *Hamster* | 16541.74 | 8152.74 | 6965.82 | 104166.54 |
| *USFCA72* | 36135.65 | 15550.61 | 12256.00 | 214740.29 |
| *NIPS-Ego* | 29764.71 | 14175.33 | 14354.80 | 147793.07 |

Fix any feasible defense seed set $S_B$. By Eq. (4), $\Phi_{\mathcal{A}}(S_B) = \max_{S_A \in \mathcal{A}} F(S_A, S_B)$. Since $\pi$ is a distribution over $\mathcal{A}$, we have

$$\Phi_{\mathcal{A}}(S_B) \geq \sum_{i=1}^{p} \pi_i F(S_A^i, S_B).$$

By Lemma 6, each term satisfies $F(S_A^i, S_B) \geq h_i(S_B)$. Therefore,

$$\Phi_{\mathcal{A}}(S_B) \geq \sum_{i=1}^{p} \pi_i h_i(S_B) = h_\pi(S_B) = K_\pi^B - \Psi_\pi^B(S_B).$$

Taking the minimum over all feasible defenses gives

$$\mathrm{OPT}_{\mathcal{A}} \geq K_\pi^B - \max_{S_B \in \mathcal{S}_B} \Psi_\pi^B(S_B).$$

By the definition of $\Psi_{\max,\pi}^B$, we have

$$\max_{S_B \in \mathcal{S}_B} \Psi_\pi^B(S_B) = \Psi_{\max,\pi}^B.$$

Combining this equality with Eq. (24) gives

$$\mathrm{OPT}_{\mathcal{A}} \geq K_\pi^B - \frac{\widehat{\Psi}_\pi^B(S_B^\circ(\pi))}{1 - \frac{1}{e}} - \varepsilon_B = L_{\mathrm{mix}}(\pi).$$

The inequality $\mathrm{OPT}_{\mathcal{A}} \leq \mathrm{OPT}$ follows because $\mathcal{A}$ is a subset of the full feasible attack family. This proves $L_{\mathrm{mix}}(\pi) \leq \mathrm{OPT}_{\mathcal{A}} \leq \mathrm{OPT}$. □

*Proof of Lemma 10.* Algorithm 4 sets $L = L_{\mathrm{mix}}(\pi)$ for the uniform distribution $\pi$ over the attack pool. By Lemma 9, $L_{\mathrm{mix}}(\pi) \leq \mathrm{OPT}_{\mathcal{A}} \leq \mathrm{OPT}$. Therefore, $L \leq \mathrm{OPT}$. □

*Proof of Theorem 11.* Let $\mathcal{E}_t^A$ and $\mathcal{E}_t^B$ denote the events that the attack-side upper certificate and the defender-side lower certificate are valid at iteration $t$, respectively. By Lemmas 5 and 10, the failure probabilities of these events are at most $\delta_t^A$ and $\delta_t^B$, respectively. Since Algorithm 5 sets $\delta_t^A = \delta_t^B = \delta/[2t(t+1)]$, the union bound gives a total failure probability of at most $\sum_{t=1}^{\infty}(\delta_t^A + \delta_t^B) = \delta \sum_{t=1}^{\infty} 1/[t(t+1)] = \delta$. Thus, with probability at least $1-\delta$, all certificates generated during the execution of Algorithm 5 are simultaneously valid.

Condition on this event. For every executed iteration $t$, the defense-side certificate gives $L_t \leq \mathrm{OPT}$, and the attack-side certificate gives $\Phi(S_B^t) \leq U_t$. Since $S_B^t$ is a feasible defense, $\mathrm{OPT} \leq \Phi(S_B^t)$. Therefore, $L_t \leq \mathrm{OPT} \leq \Phi(S_B^t) \leq U_t$. If Algorithm 5 terminates with $U_t - L_t \leq \varepsilon$, then $\Phi(S_B^t) - \mathrm{OPT} \leq U_t - L_t \leq \varepsilon$, which proves Eq. (18). □

## APPENDIX C
## ADDITIONAL EXPERIMENTAL RESULTS

This section provides additional results complementing the experimental evaluation in Section V.

### A. Additional Running-Time Results under the Default Setting

Table VI reports the running time of the compared methods under the default setting. *CRD* requires more running time than *FJ-Iter* and *HIST-Iter*, whereas *CRD*$^+$ completes all evaluated larger datasets, including *Orkut*. A dash indicates that the corresponding method is not evaluated on that dataset due to excessive running time.

### B. Additional Results on Asymmetric Seed Budgets

We examine asymmetric attack and defense seed budgets by varying $k_A$ and $k_B$ separately around the default setting. Fig. 4 shows that increasing the attack budget generally leads to higher final polarization, whereas increasing the defense budget reduces it. The relative performance of the compared

methods remains broadly consistent with the main results. These observations are consistent with the opposing roles of the attack and defense cascades.

### C. Additional Results on Cascade Parameters

For simplicity, we set $\rho_u = \rho$ for all $u \in V$ and vary $\rho$, which controls how much defensive exposure suppresses the adversarial effect. As shown in Fig. 5, the relative performance of the compared methods remains consistent with the main results. The final polarization generally decreases as $\rho$ increases, which is consistent with the stronger suppression of the adversarial effect.

Fig. 6 examines different combinations of the cascade strengths $\alpha_A$ and $\alpha_B$. The relative performance of the compared methods remains broadly consistent across different settings. Increasing $\alpha_A$ generally increases polarization, whereas increasing $\alpha_B$ reduces it, which is consistent with the respective effects of the polarizing and depolarizing cascades.

### D. Additional Results on the Iteration Limit

Fig. 7 reports the final polarization under different maximum numbers of outer iterations $T$. On most datasets, increasing $T$ has only a limited effect on the final polarization. This result supports $T = 10$ as a reasonable default setting in our experiments.

### E. Additional Results on the Sampling Budget

Table VII reports the final polarization when varying the sampling budget from $10^3$ to $10^6$. Increasing the sampling budget generally improves or stabilizes the final polarization, while the improvement becomes limited beyond $10^5$ on most datasets. Table VIII shows that larger sampling budgets substantially increase the running time. Overall, these results support $10^5$ as a practical default that balances solution quality and computational cost.

### F. Additional Results on the Certificate Tolerance

Table IX reports the final polarization under different relative certificate tolerances $\varepsilon/n$. Across the tested datasets, varying $\varepsilon/n$ from $2\%$ to $10\%$ has only a limited effect on the final polarization. In contrast, Table X shows that the running time decreases substantially as the tolerance is relaxed. These results illustrate the trade-off between certificate tolerance and computational cost, while the resulting solution quality remains relatively stable.

### G. Additional Results under Disagreement

Table XI reports the defense effectiveness when disagreement is used as the conflict objective. The overall performance pattern remains consistent with the main results. *CRD* remains competitive on the datasets where it is evaluated, while *CRD*$^+$ achieves the lowest final disagreement on all larger datasets. These results show that the effectiveness of the proposed methods is not limited to polarization and extends to disagreement.

### H. Additional Results under the LT Model

Table XII reports the defense effectiveness under the LT model. The overall performance pattern remains consistent with the results under the IC model. *CRD* remains competitive on the datasets where it is evaluated, while *CRD*$^+$ achieves the lowest final polarization on all larger datasets. These results demonstrate that the proposed methods remain effective under a different diffusion model.

### I. Additional Results with Exponential Innate Opinions

To examine the effect of the innate-opinion distribution, we replace the default uniform opinions with fixed exponential-style opinion vectors. For each node $u$, we first sample $U_u$ uniformly from $[0, 1]$, set $y_u = 1 - \log(1 - U_u)$, and then normalize the values as $s_u = y_u / \max_{v \in V} y_v$. Table XIII reports the corresponding defense effectiveness. The overall performance pattern remains broadly consistent with the main results. *CRD* remains competitive on the datasets where it is evaluated, while *CRD*$^+$ achieves the lowest final polarization on all larger datasets. These results indicate that the proposed methods remain effective under a different innate-opinion distribution.

### J. Additional Running-Time Results for the Ablation Study

Table XIV reports the running times of *CRD* and its ablation variants. *CRD*$'$ consistently reduces the running time by retaining only the most recently generated attack, while *CRD*$''$ generally incurs lower computational cost by removing the empirical defense search. In contrast, *CRD*$^\dagger$ requires substantially more running time across all datasets considered, reflecting the additional cost of adaptively updating the mixed-attack distribution. These results further illustrate the computational trade-offs among the main components of *CRD*.